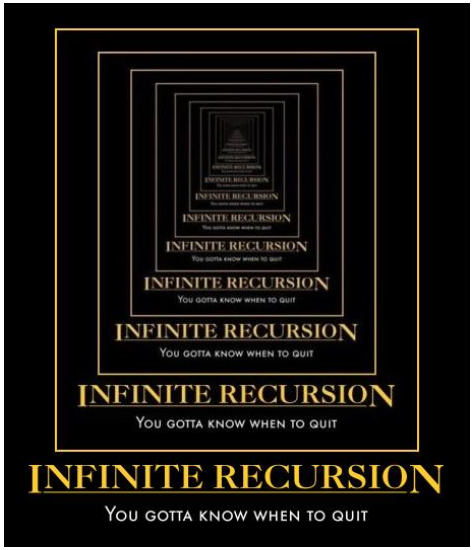
COMP/EECE 7/8745 Neural Networks

Topics:

- Limitations of DNN or CNN architectures
- Why Recurrent Neural Networks (RNN)?
- What is Recurrent Neural Networks (RNN) model
- Learning with RNN models
- Where to apply RNN (applications)

Md Zahangir Alom
Department of Computer Science
University of Memphis, TN

New Topic: RNNs



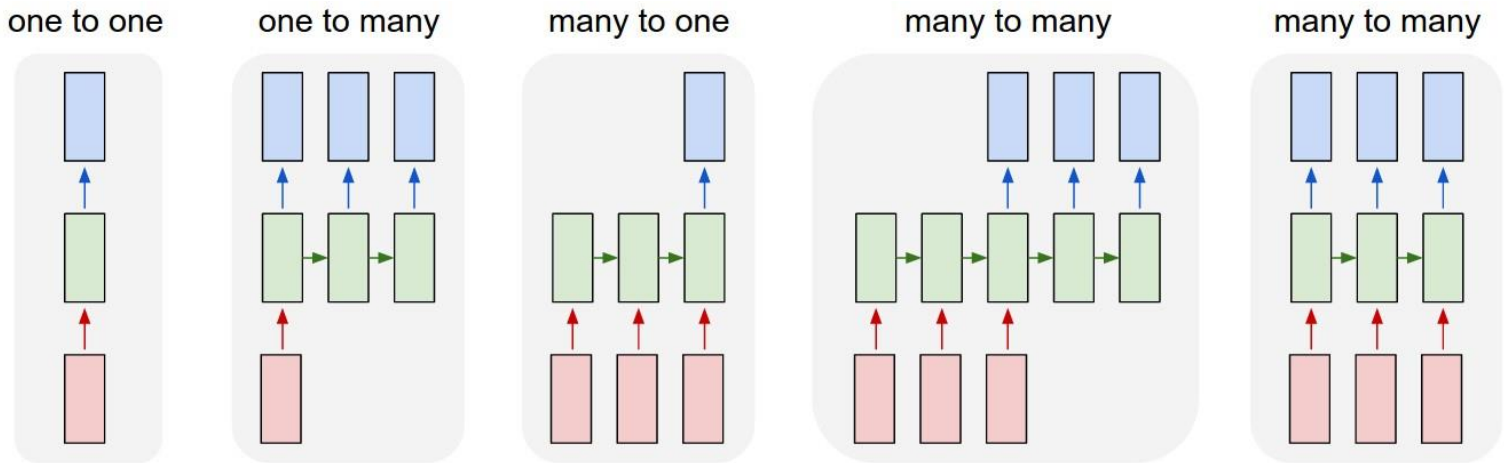


Image Credit: Andrej Karpathy

New Terms

- Recurrent Neural Networks (RNNs)
- Recursive Neural Networks
  - General family; **think graphs instead of chains**
- Types:
  - **“Vanilla” RNNs (Elman Networks)**
  - Long Short-Term Memory (LSTMs)
  - Gated Recurrent Units (GRUs)
  - Variants
- Optimization algorithms
  - BackProp Through Time (BPTT)
  - BackProp Through Structure (BPTS)

Limitations of DNN or CNN architectures

- **Problem 1**: Can't model sequences
  - Fixed-sized Inputs & Outputs
  - No temporal structure

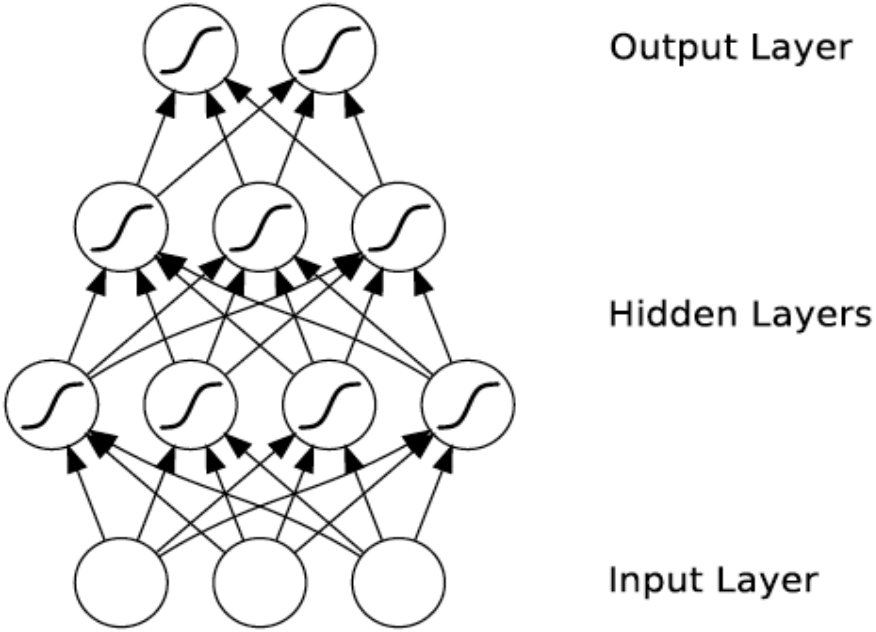


Image Credit: Alex Graves, book

Limitations of DNN or CNN architectures

- **Problem 2:** Pure feed-forward processing
  - No “memory”, no feedback

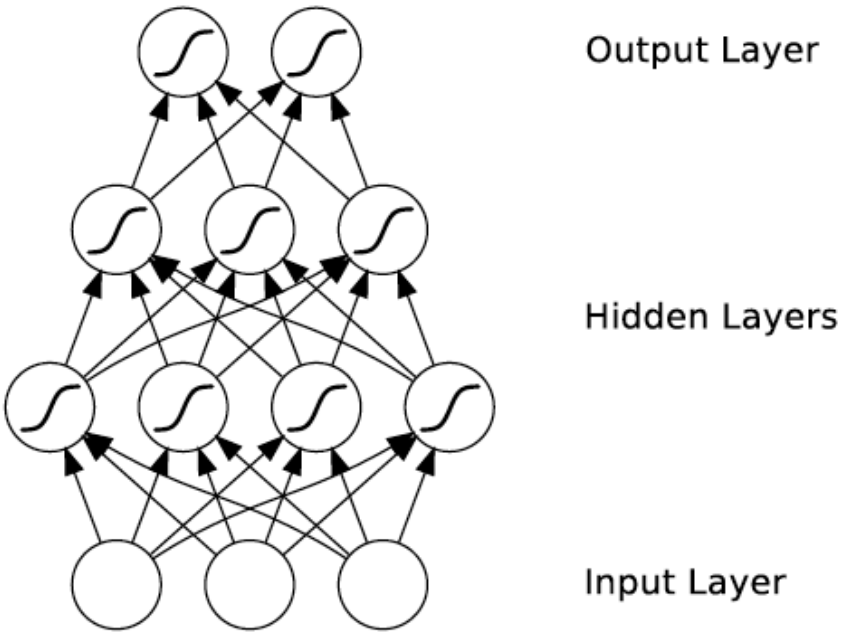


Image Credit: Alex Graves, book

Why Recurrent Neural Networks (RNNs)?

- **Human** don’t throw everything away and start thinking from scratch again.
  - As you read this essay, you **understand each word based on your understanding of previous words**.

- Traditional neural network approaches **cannot solve this type of problems**, RNN address this issues.

Why Recurrent Neural Networks (RNNs)?

- Not all problems can be converted into **one with fixed-length inputs and outputs**
  - Problems such as Speech Recognition or Time-series prediction **require a system to store and use context information**
    - Simple case: Output YES if the number of 1s is even, else NO

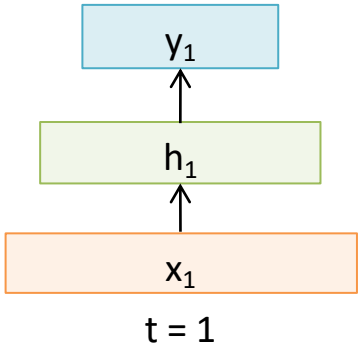
1000010101 – YES, 100011 – NO, …

- Hard/Impossible **to choose a fixed context window**
  - There can always be a **new sample longer or shorter than anything seen**

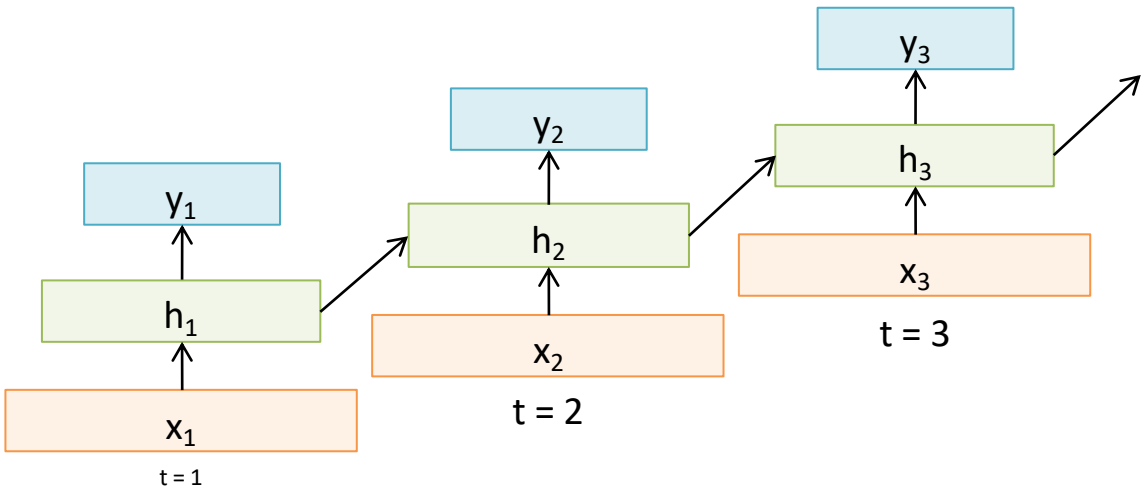
What is Recurrent Neural Network?

- **RNN** is a type of neural network designed to process **sequential data** by maintaining a **memory of past inputs** through its internal hidden state.
- **What does that mean in simple terms?**
  - Unlike standard feedforward neural networks, an RNN:
    - **Reuses information from previous time steps**
    - Maintains a **hidden state** that captures **historical context**
    - Processes inputs **one step at a time**, making it ideal for sequences

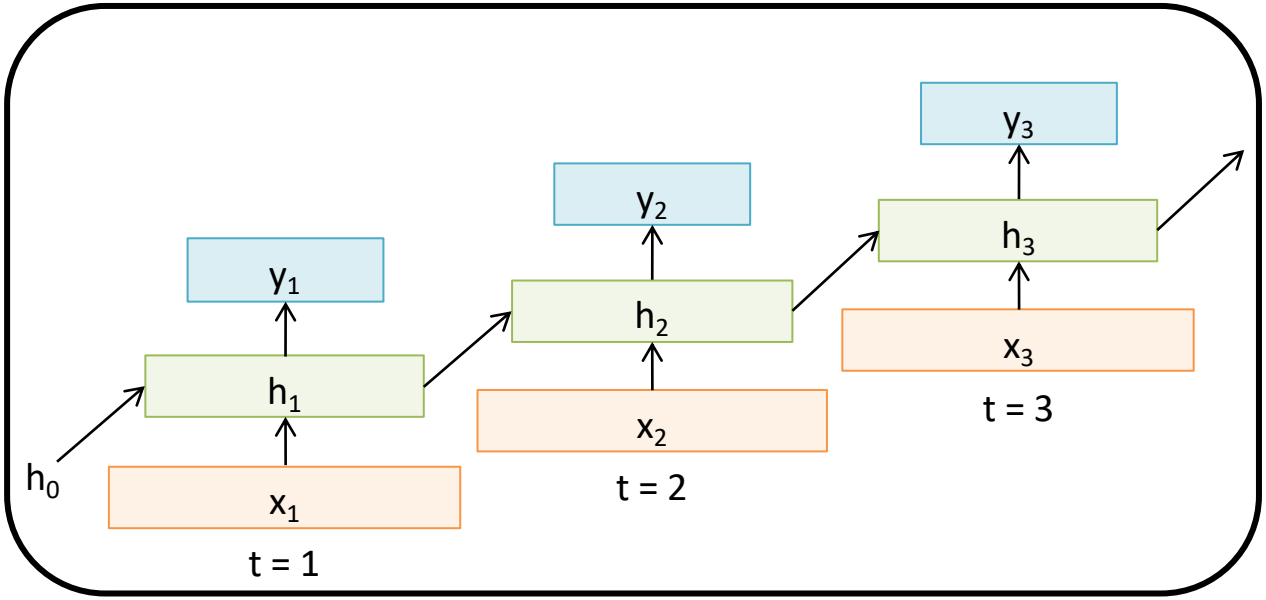
Sample Feed-forward Network (FFN)

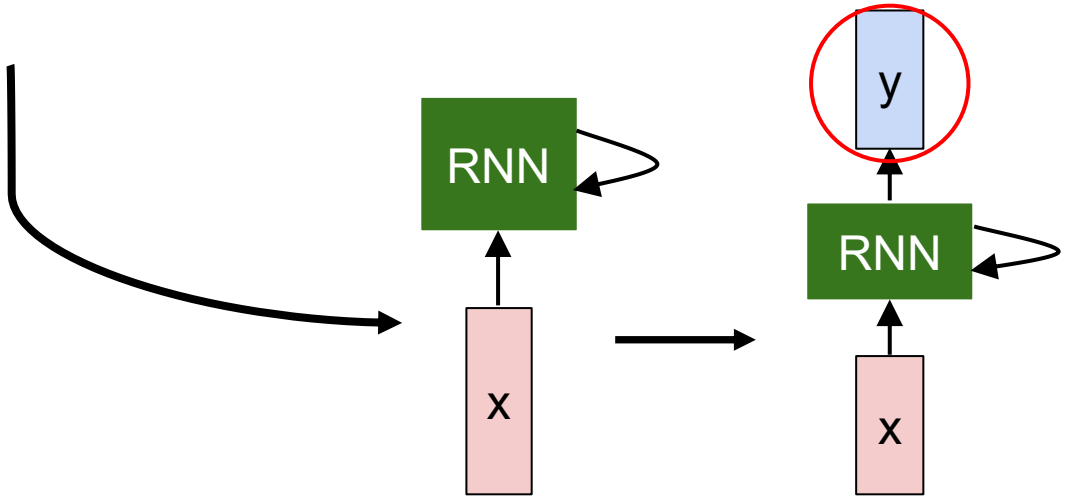


FFN to Sample RNN



FFN to Sample RNN Structural





usually want to predict a vector at some time steps

Mathematical model of RNN

We can process a sequence of vectors **x** by applying a **recurrence formula** at every time step:

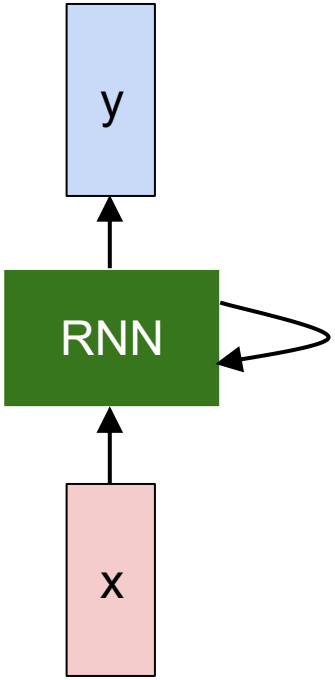
$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

new state

old state

input vector at some time step

some function with parameters W



*At time step **t***, an RNN takes:

- The **current input** x_t
- The **previous hidden state** h_{t-1}

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

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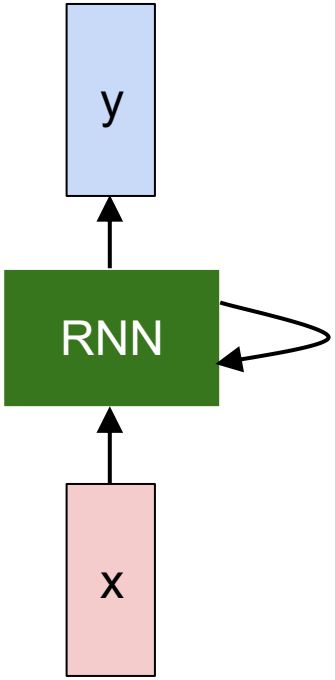
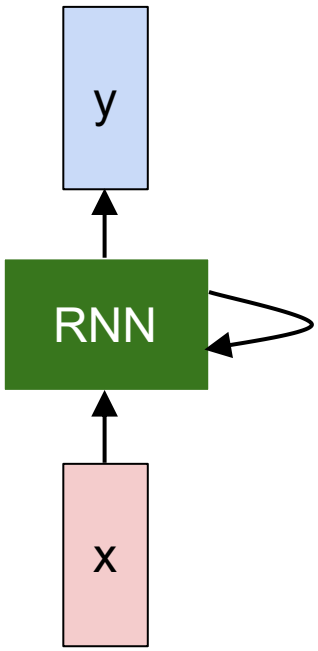


Notice: the **same function and the same set of parameters** are used at every time step.

Mathematical model of RNN

The state consists of a single *"hidden"* vector **h**:



$$h_t = f_W(h_{t-1}, x_t)$$

$$\downarrow$$

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h)$$

$$\downarrow$$

$$y_t = W_{hy}h_t + b_y$$

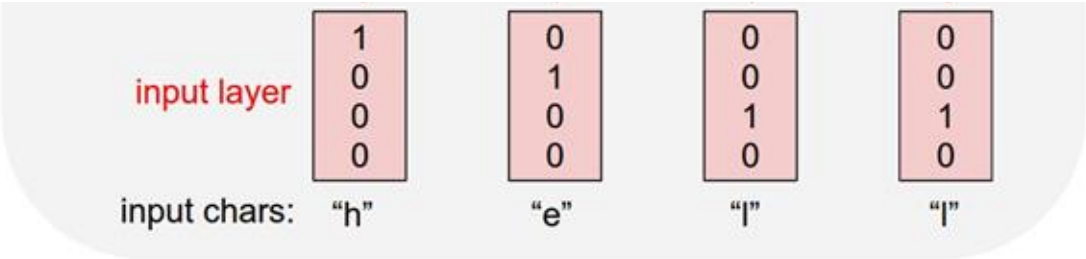
Sometimes called a "Vanilla RNN" or an "Elman RNN" after Prof. Jeffrey Elman

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Example: Character-level Language Model

Vocabulary:
[h,e,l,o]

Example training sequence:
“hello”



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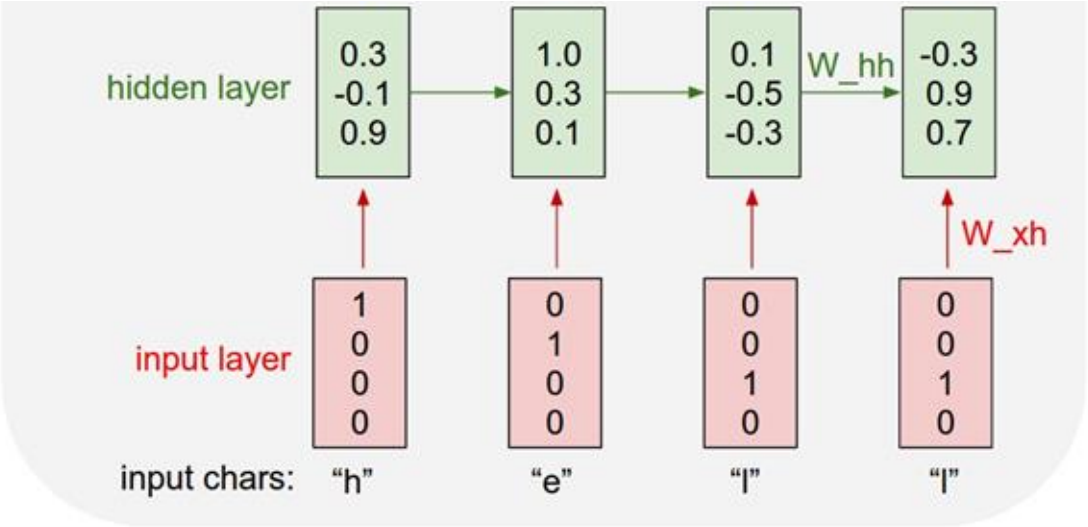

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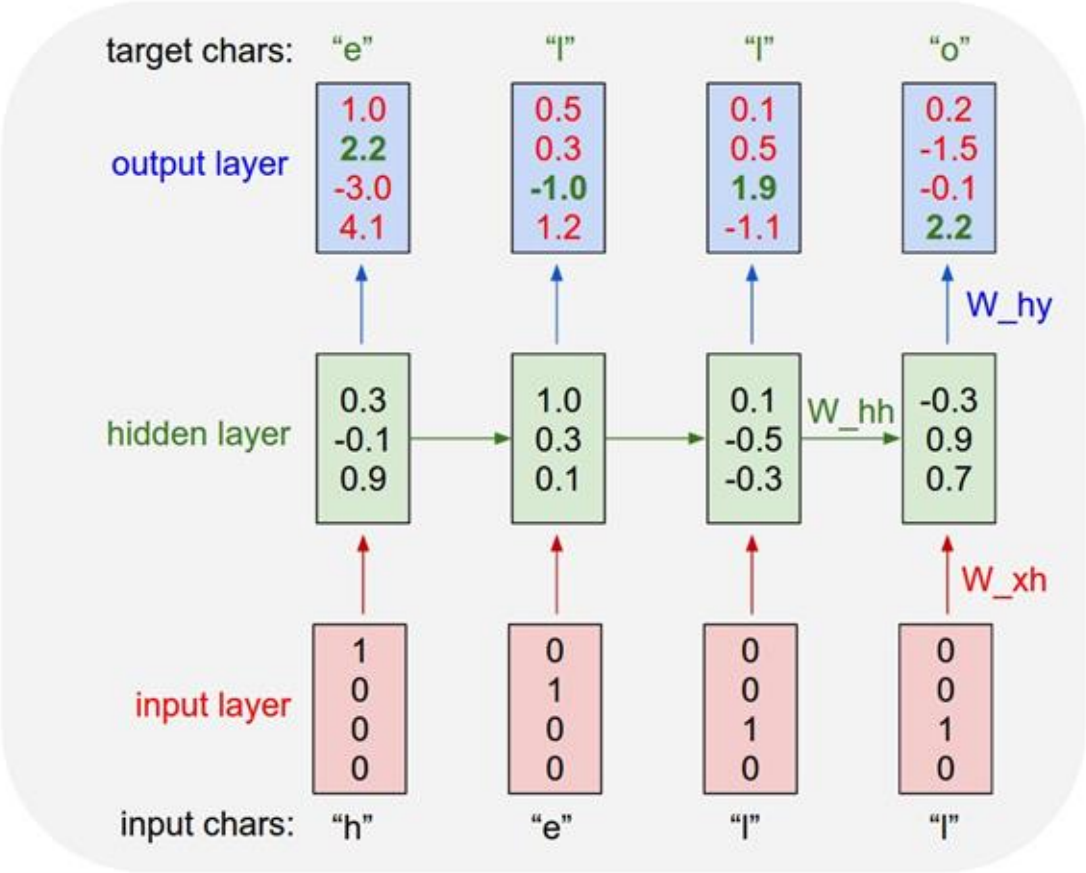

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Training Time: “Teacher Forcing”

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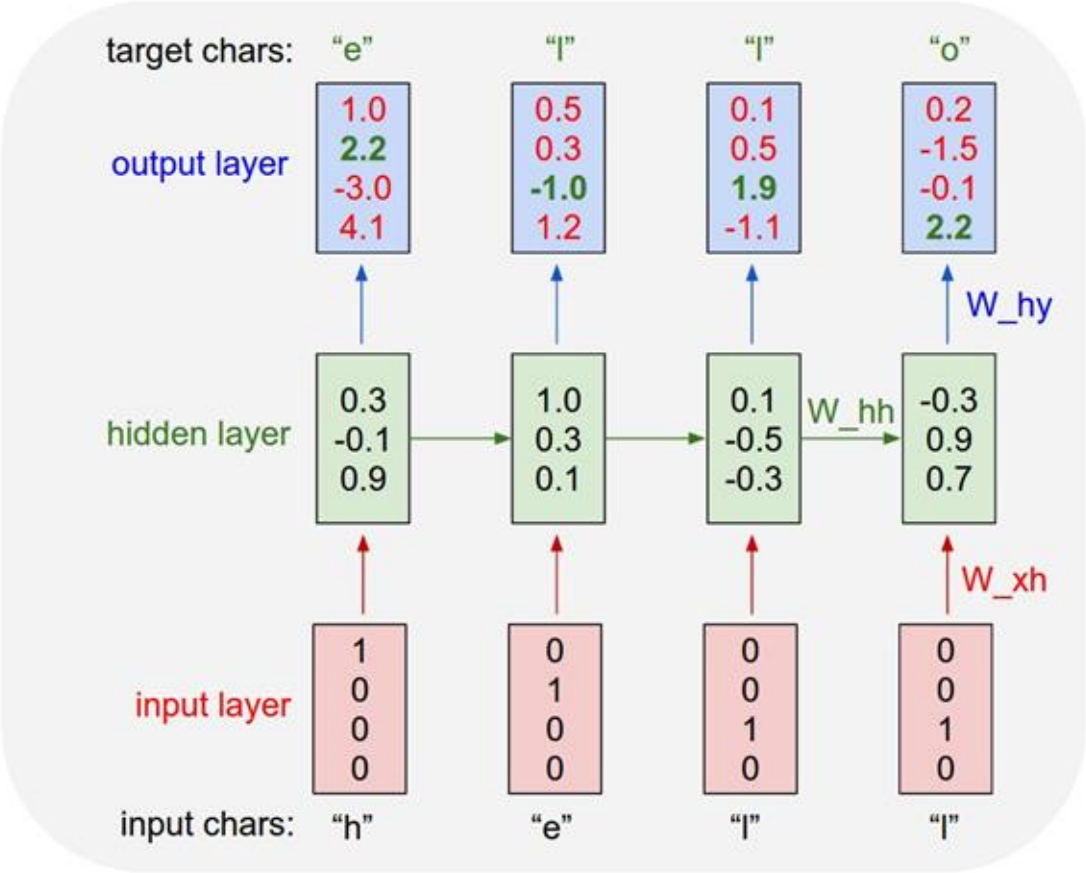

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Test Time: Sample / Argmax / Beam Search

Example: Character-level Language Model Sampling

Vocabulary:
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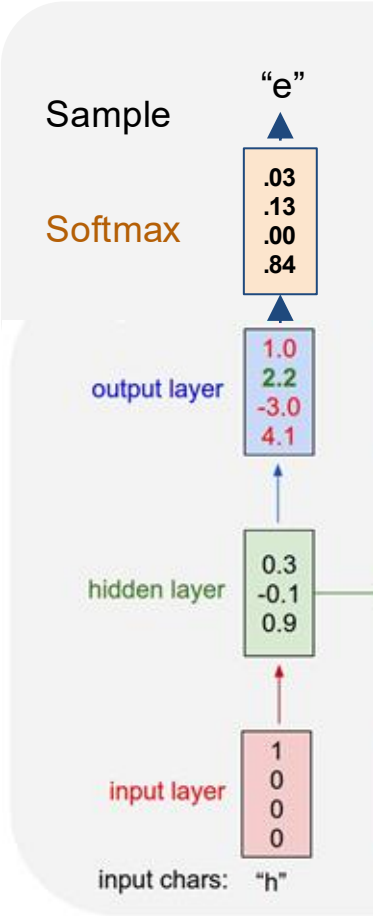
At test-time sample characters one at a time, feed back to model



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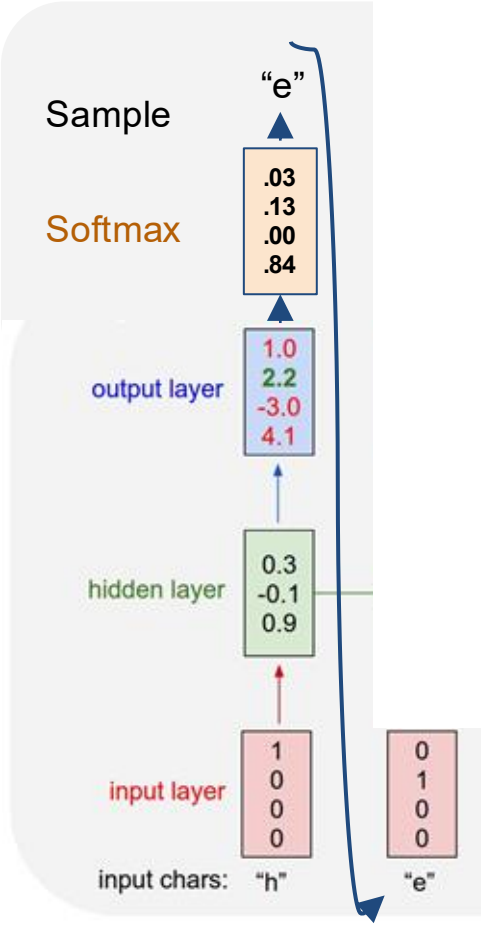

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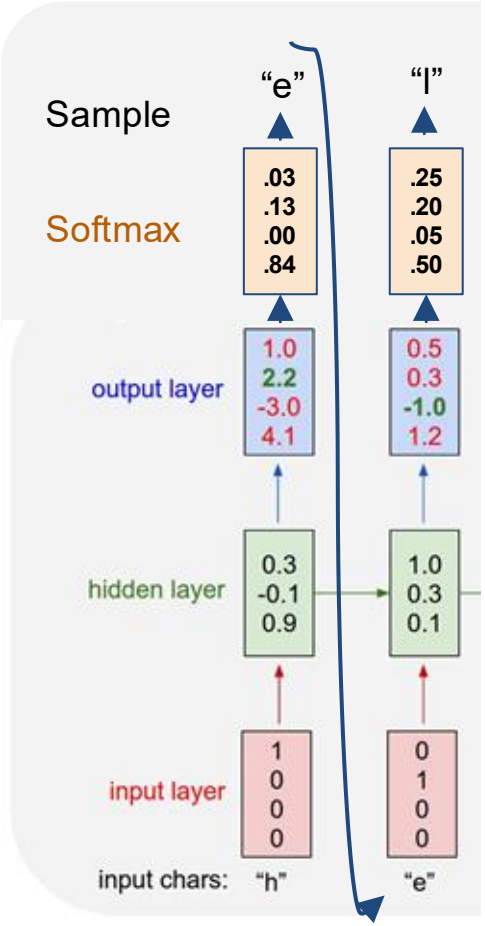

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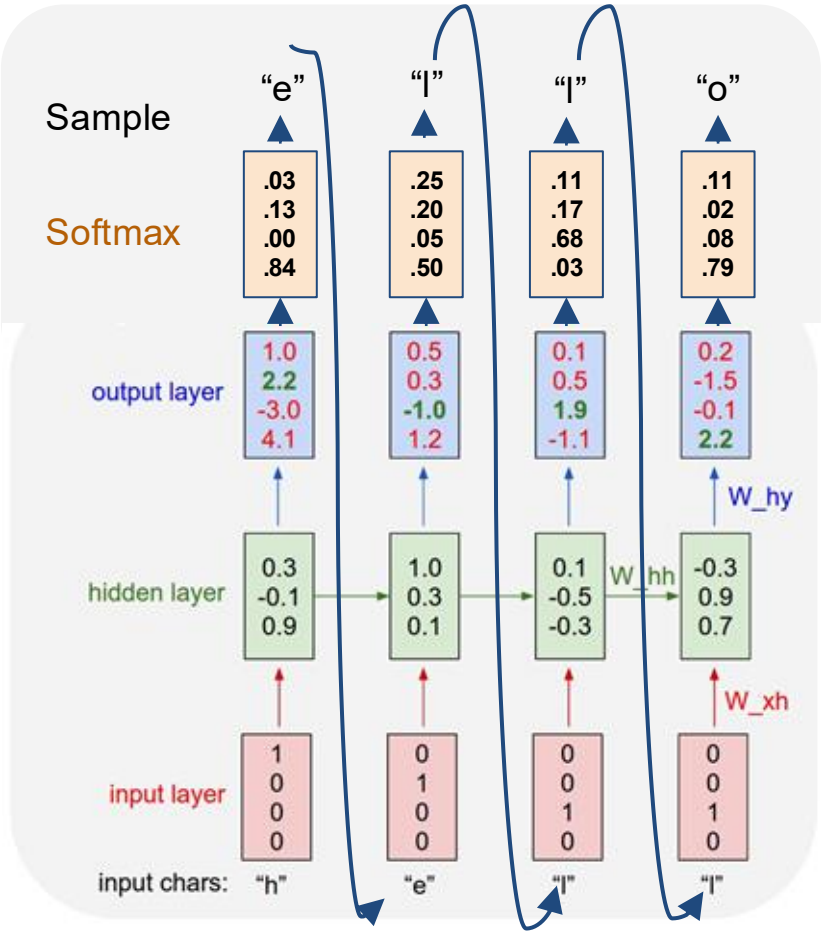
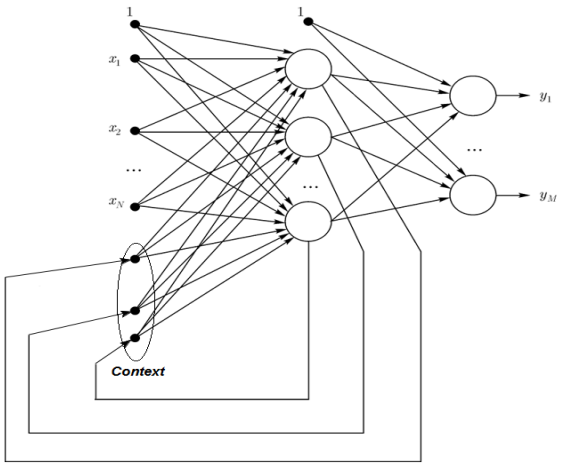
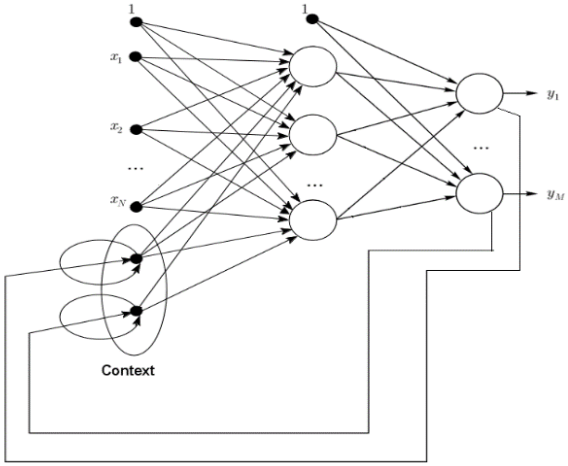


Type of RNN architectures

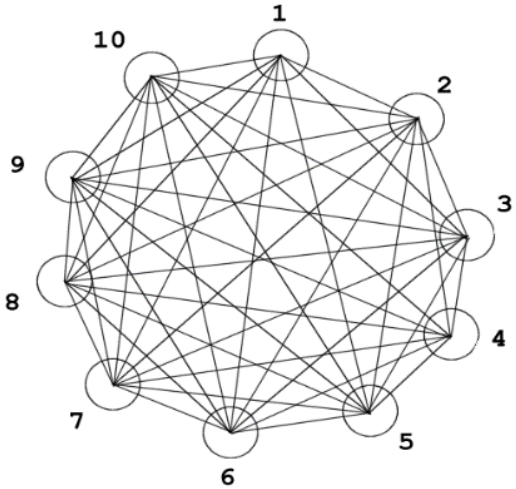
- **Three different type of RNN** architectures:
  - Elman Neural Network
  - Jordan Neural Network
  - Hopfield Neural Network



Elman Neural Network



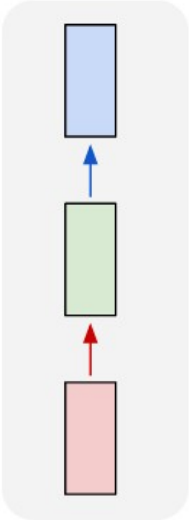
Jordan Neural Network



Hopfield Neural Network

RNNs to solve real-world problems (Input-Output)

one to one



Input: No sequence

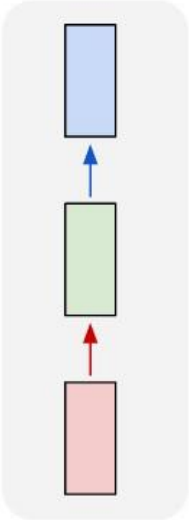
Output: No sequence

Example: “standard” classification / regression problems

Image Credit: Andrej Karpathy

RNNs to solve real-world problems (Input-Output)

one to one



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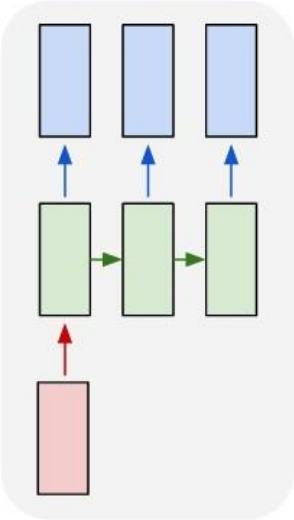
Example: “standard” classification / regression problems

one to many



Input: No sequence

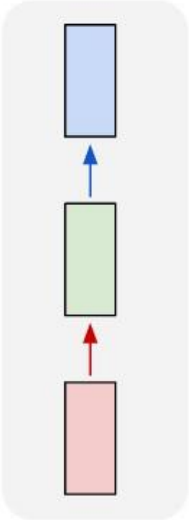
Output: Sequence

Example: Im2Caption

Image Credit: Andrej Karpathy

RNNs to solve real-world problems (Input-Output)

one to one



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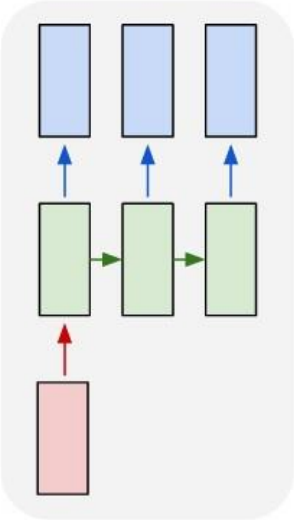
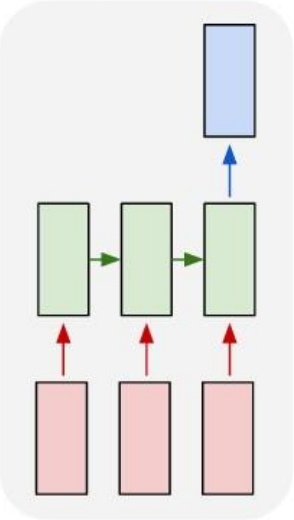


Input: No sequence

Output: Sequence

Example: Im2Caption

many to one



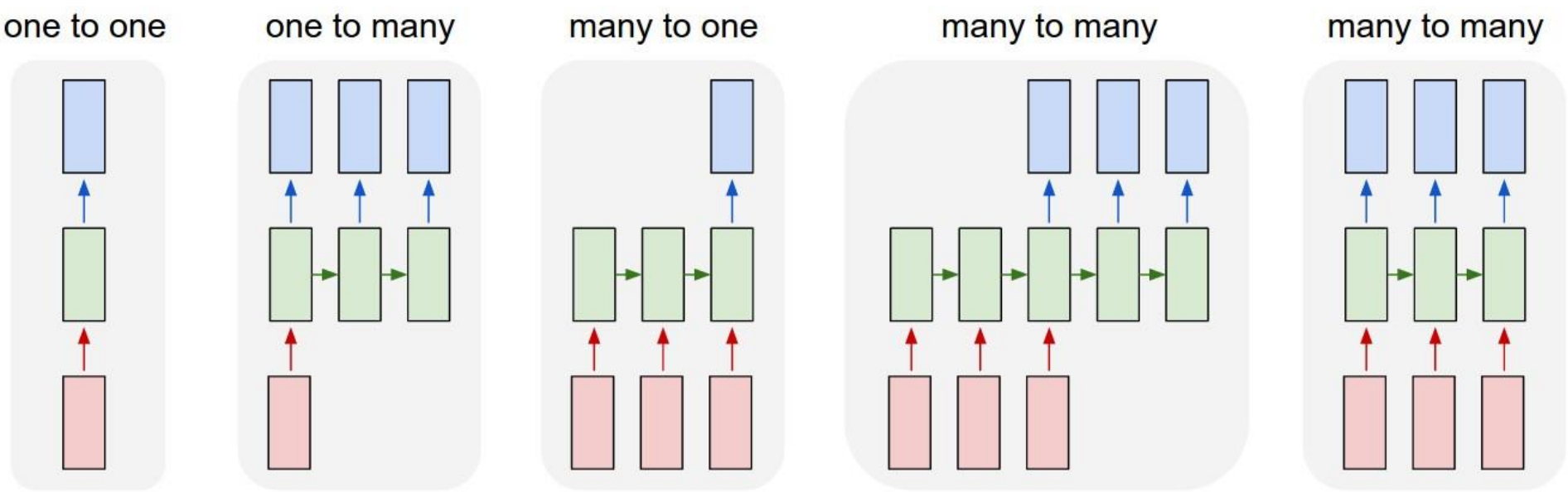
Input: Sequence

Output: No sequence

Example: sentence classification, multiple-choice question answering

Image Credit: Andrej Karpathy

RNNs to solve real-world problems (Input-Output)



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Example: “standard” classification / regression problems

Input: No sequence

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Example: Im2Caption

Input: Sequence

Output: No sequence

Example: sentence classification, multiple-choice question answering

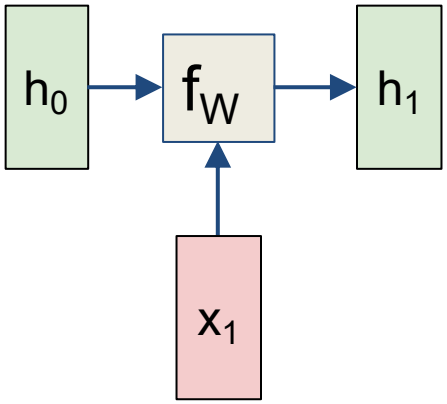
Input: Sequence

Output: Sequence

Example: machine translation, video classification, video captioning, open-ended question answering

Image Credit: Andrej Karpathy

RNN: Computational Graph



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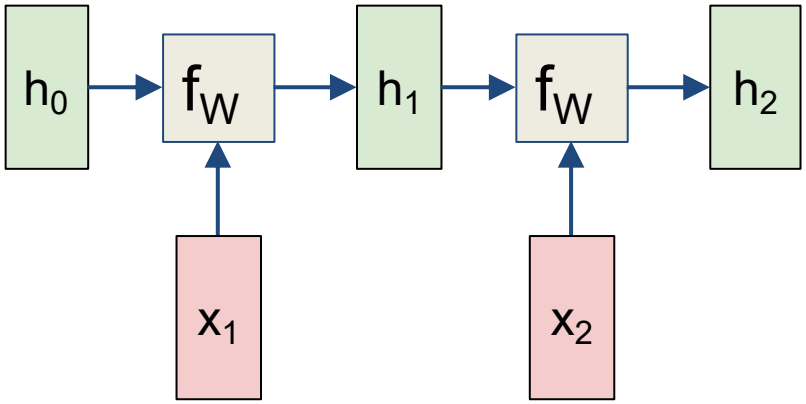

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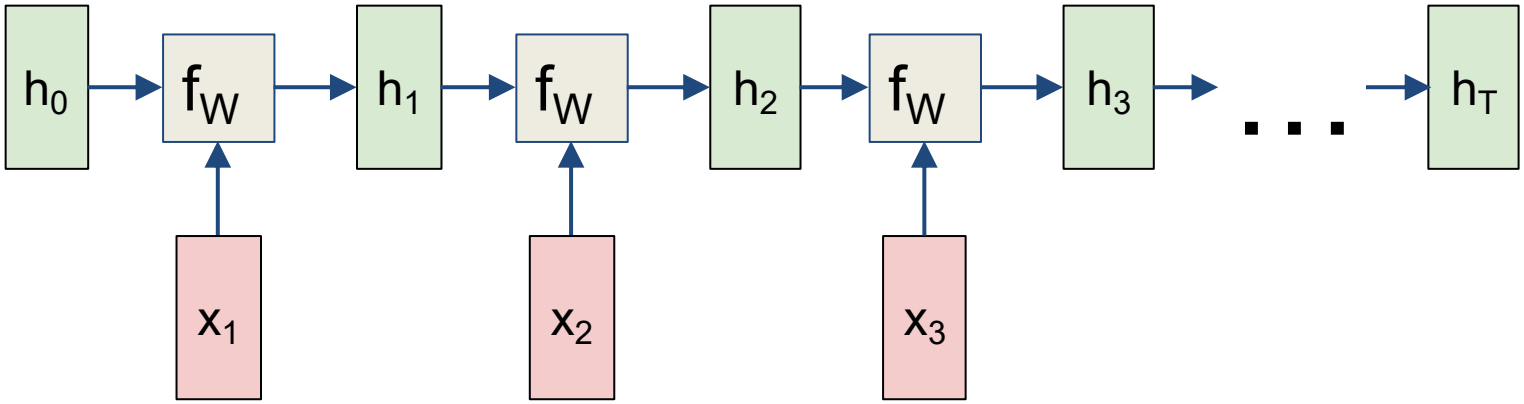

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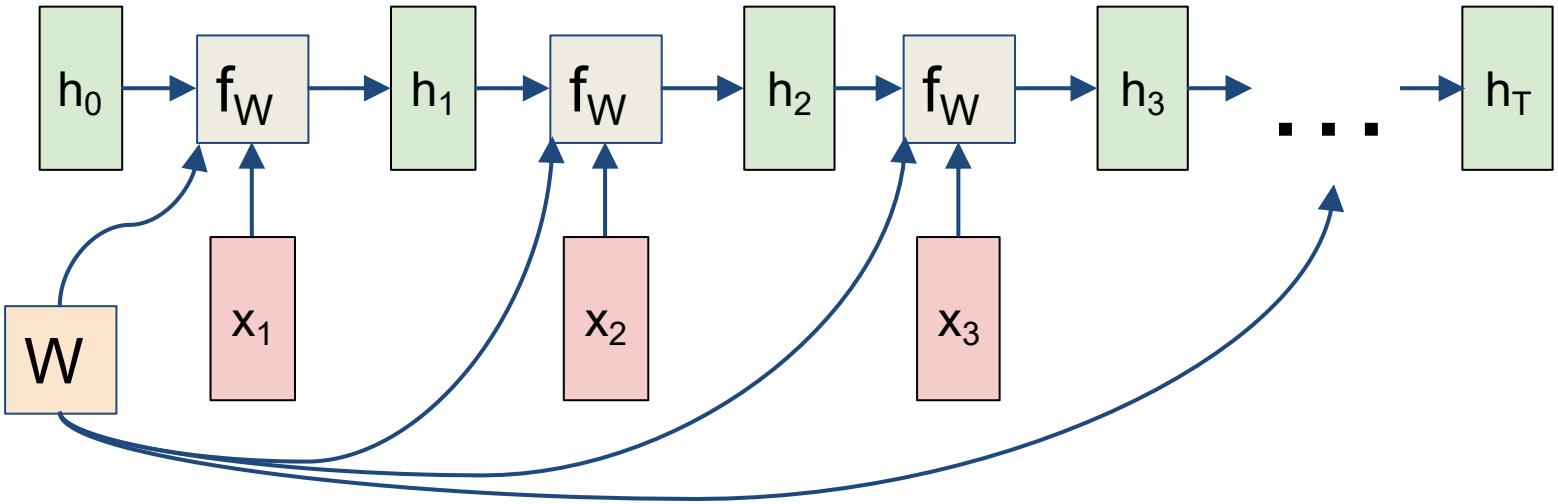
RNN: Computational Graph

Re-use the same weight matrix at every time-step



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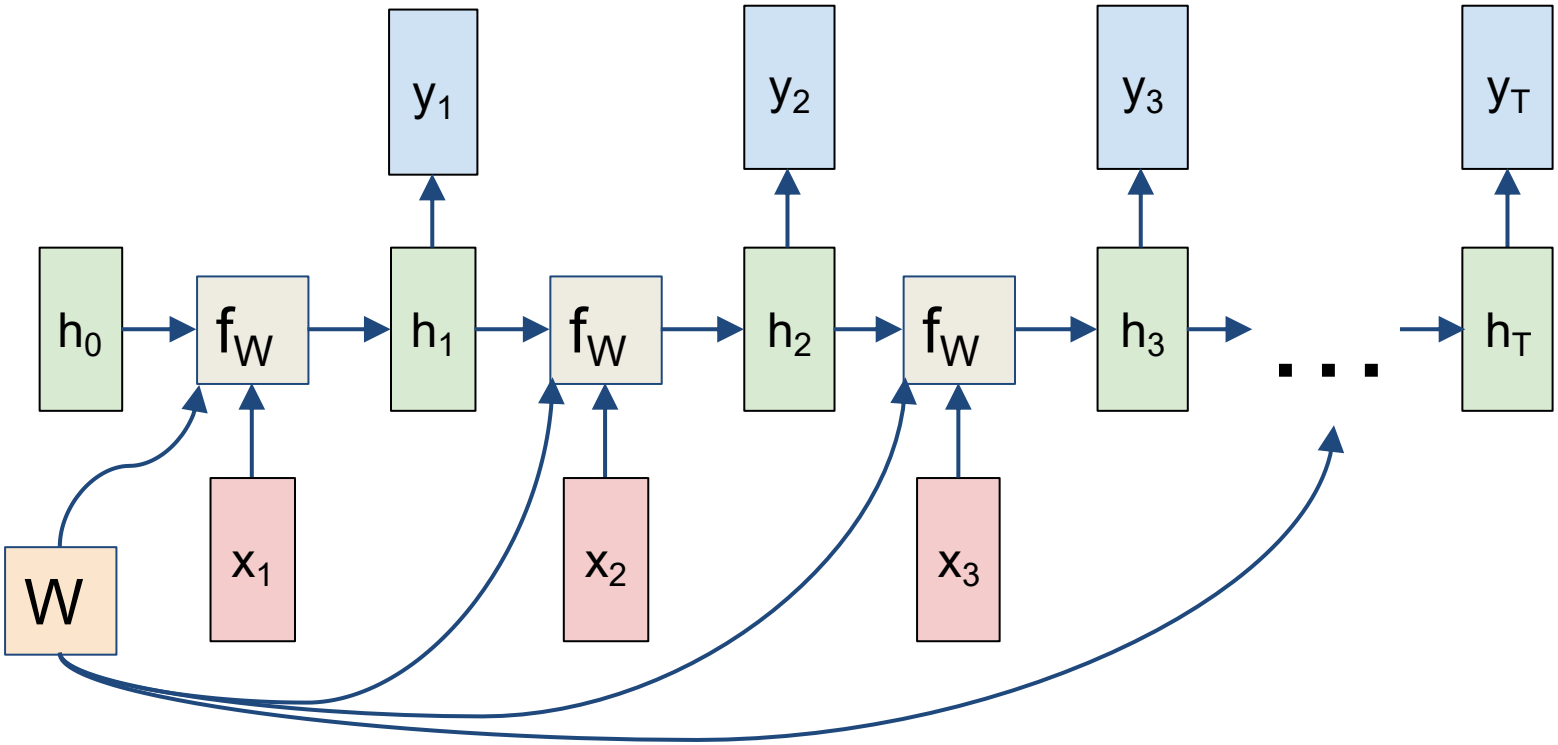
RNN: Computational Graph: Many to Many



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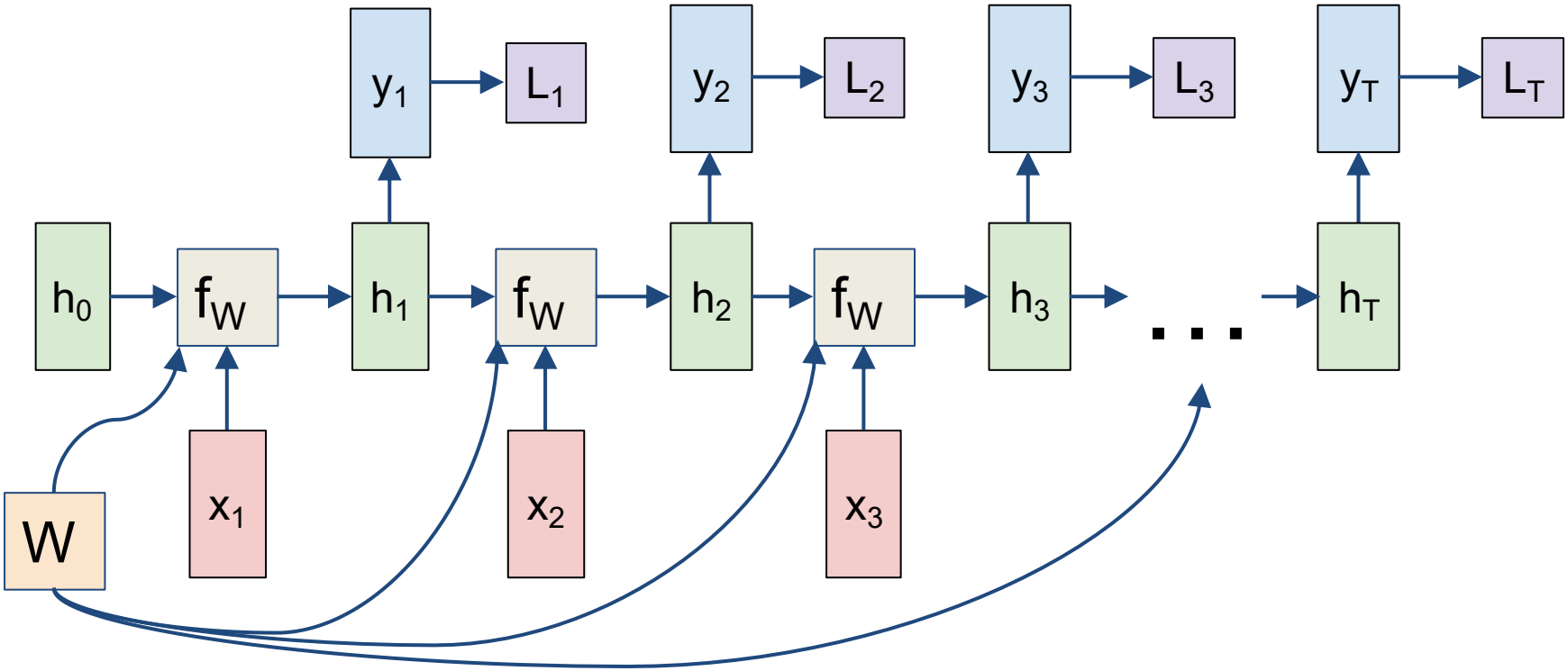

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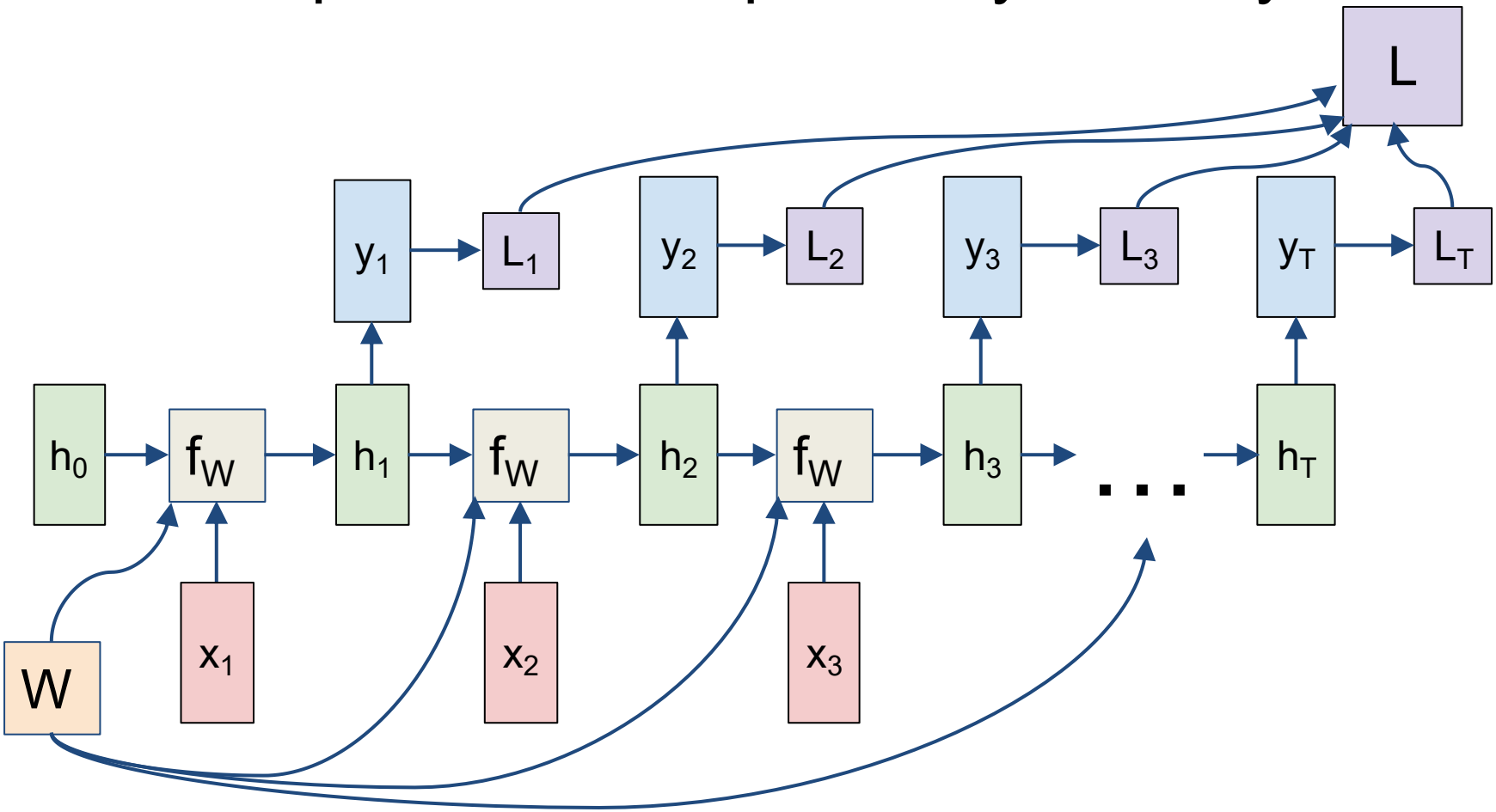

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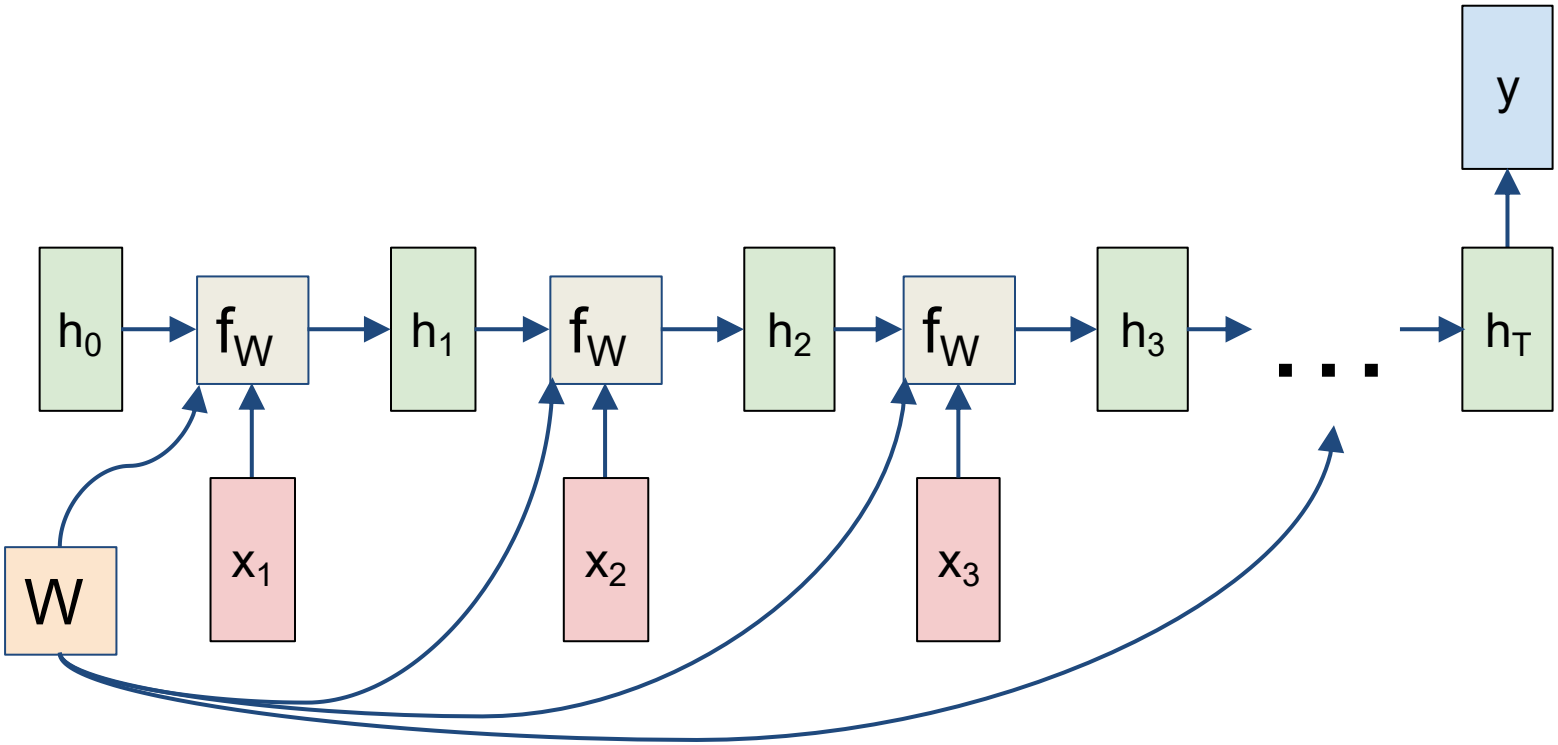
RNN: Computational Graph: Many to One



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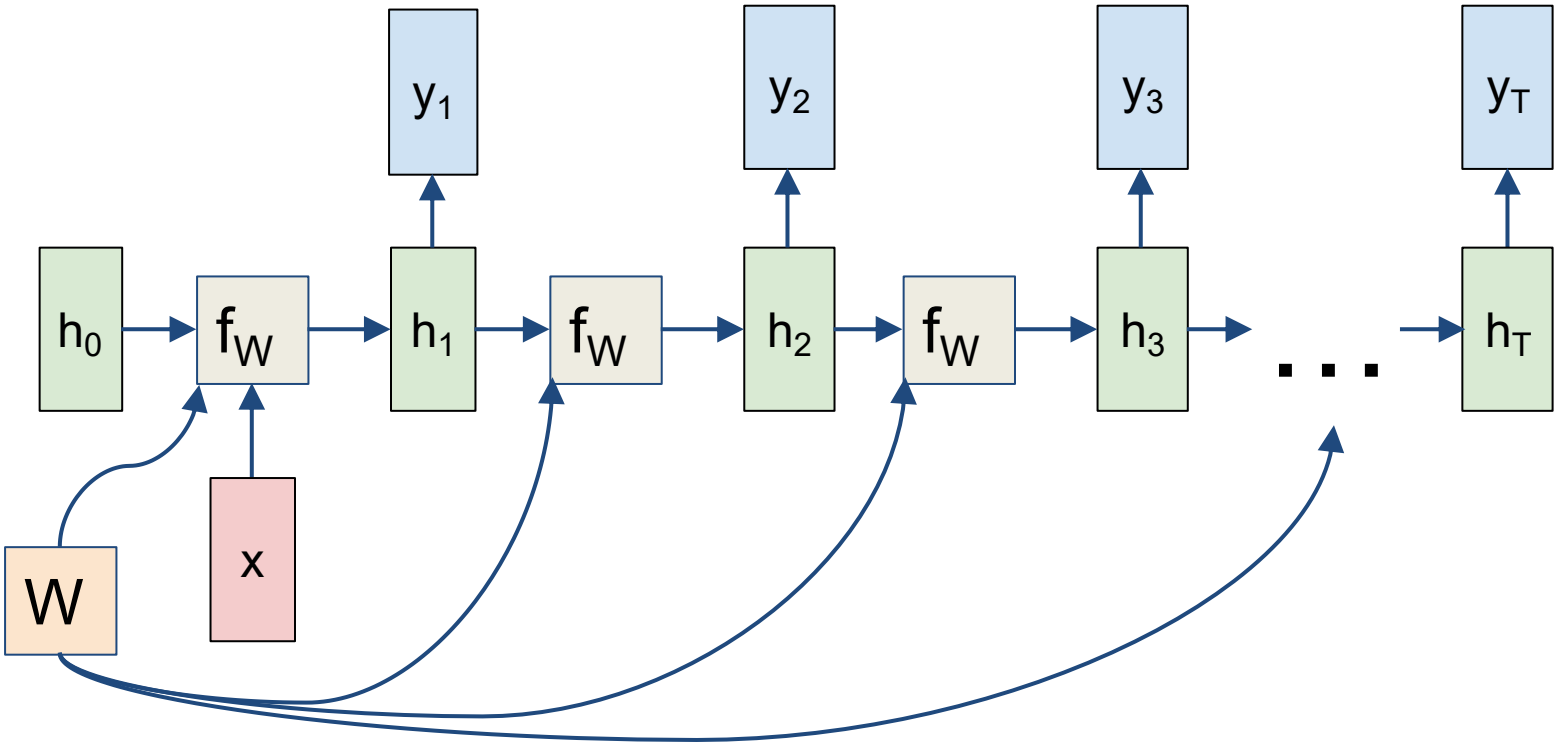
RNN: Computational Graph: One to Many



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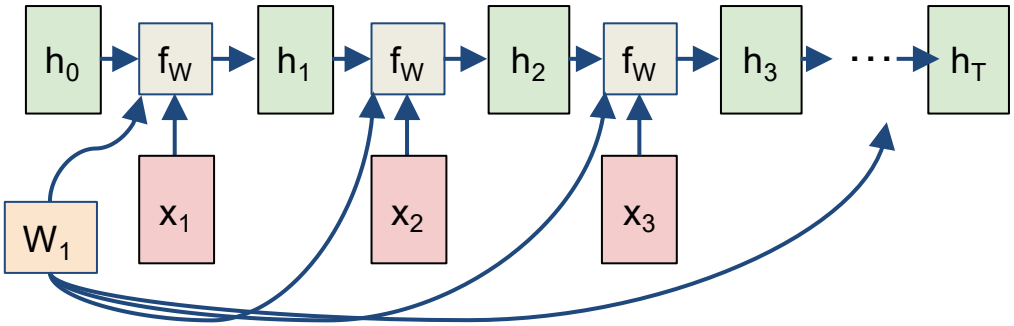
Sequence to Sequence: Many-to-one → one-to-many

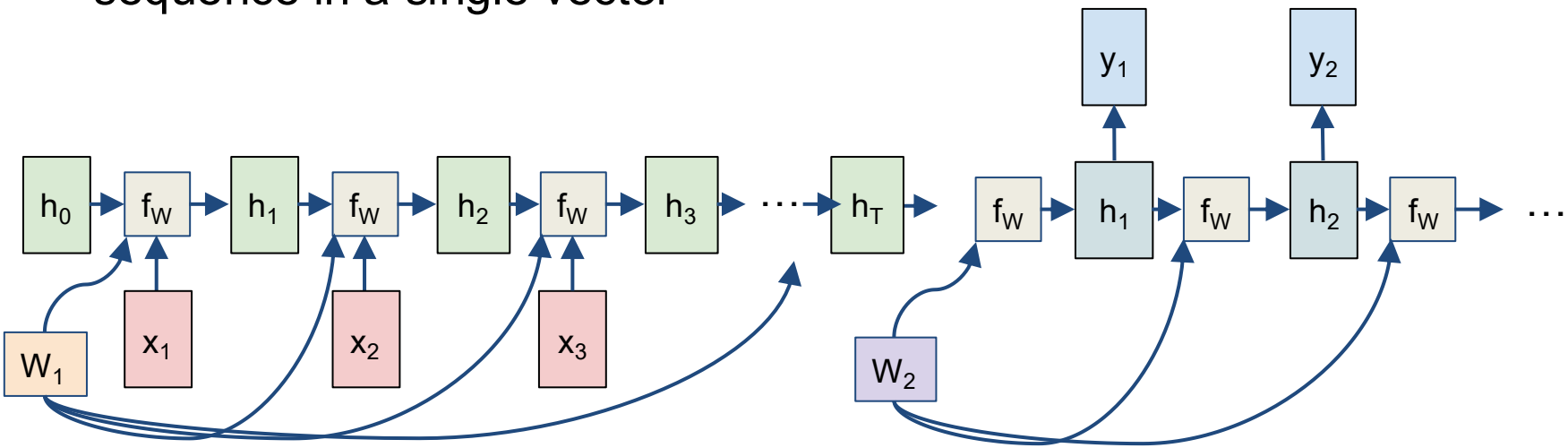
Many to one: Encode input sequence in a single vector



Sequence to Sequence: Many-to-one → one-to-many

Many to one: Encode input sequence in a single vector

One to many: Produce output sequence from single input vector

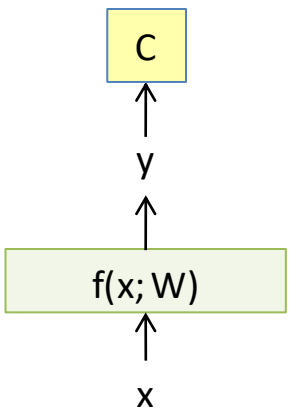


Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

2 Key Ideas

- **Parameter Sharing**
  - in computation graphs = adding gradients
- **"Unrolling"**
  - in computation graphs with parameter sharing
- Parameter sharing + Unrolling
  - Allows modeling arbitrary sequence lengths!
  - Keeps numbers of parameters in check

Chain Rule for Gradient Computation Back-Propagation for RNN (single and multiple layers)

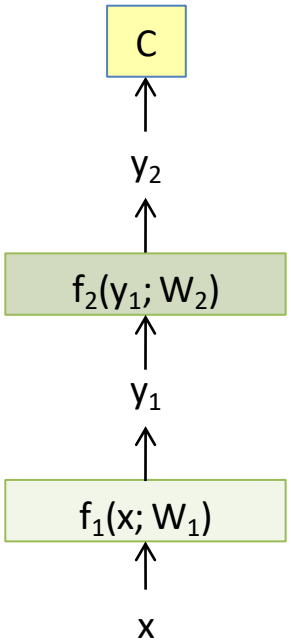


$$y = f(x;W)$$
$$C = \text{Loss}(y, y_{GT})$$

SGD Update

$$W \leftarrow W - \eta \frac{\partial C}{\partial W}$$

$$\frac{\partial C}{\partial W} = \left(\frac{\partial C}{\partial y}\right)\left(\frac{\partial y}{\partial W}\right)$$



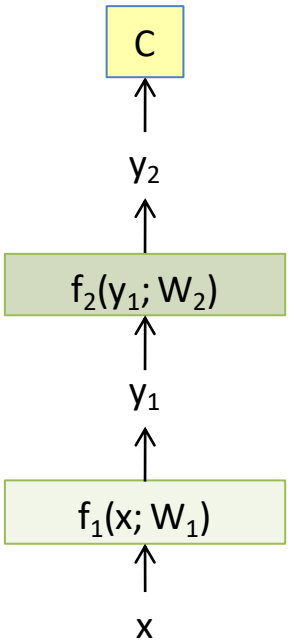
$$y_1 = f_1(x;W_1)$$
$$y_2 = f_2(y_1;W_2)$$
$$C = \text{Loss}(y_2, y_{GT})$$

SGD Update

$$W_2 \leftarrow W_2 - \eta \frac{\partial C}{\partial W_2}$$

$$W_1 \leftarrow W_1 - \eta \frac{\partial C}{\partial W_1}$$

Chain Rule for Gradient Computation



$$y_1 = f_1(x; W_1)$$

$$y_2 = f_2(y_1; W_2)$$

$$C = \text{Loss}(y_2, y_{GT})$$

$$\text{Find } \frac{\partial C}{\partial W_1}, \frac{\partial C}{\partial W_2}$$

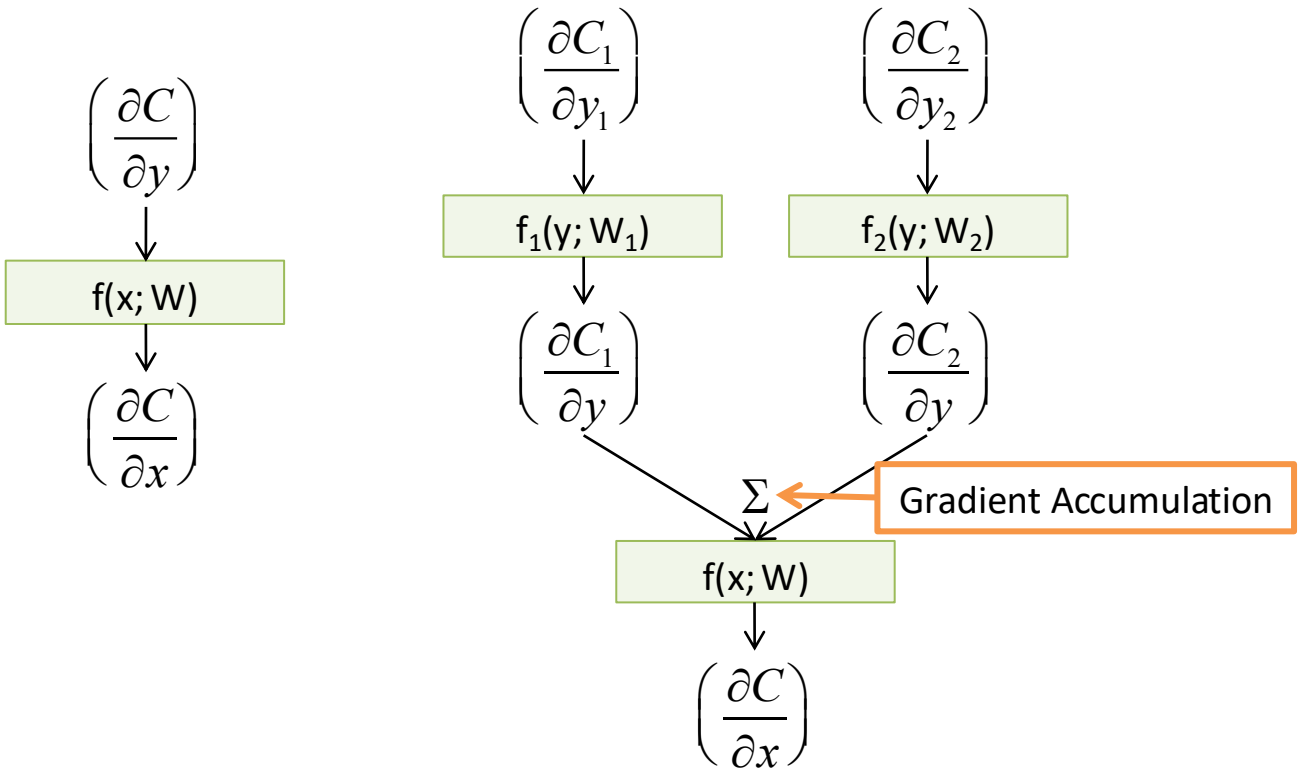
$$\frac{\partial C}{\partial W_2} = \left(\frac{\partial C}{\partial y_2}\right)\left(\frac{\partial y_2}{\partial W_2}\right)$$

$$\frac{\partial C}{\partial W_1} = \left(\frac{\partial C}{\partial y_1}\right)\left(\frac{\partial y_1}{\partial W_1}\right)$$

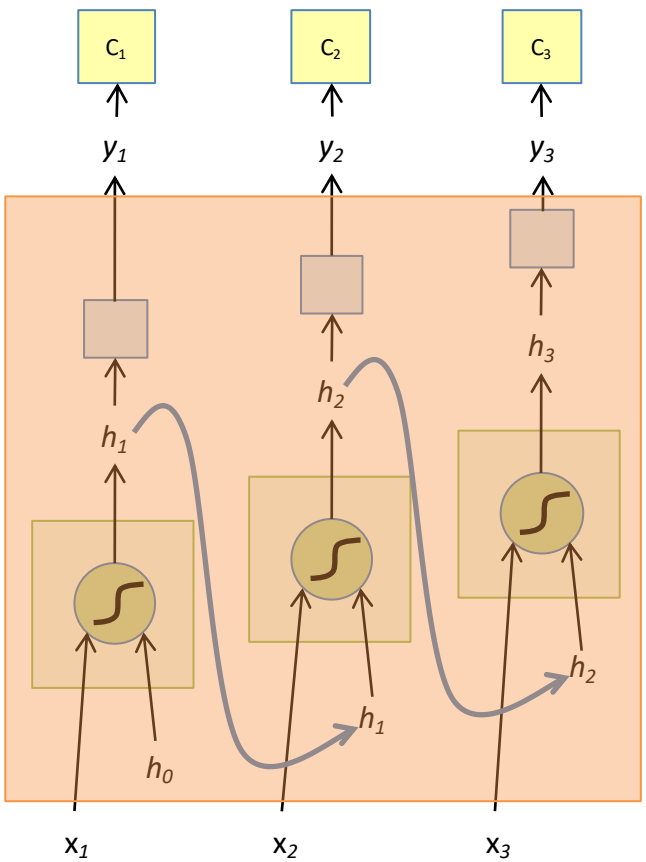
$$= \left(\frac{\partial C}{\partial y_2}\right)\left(\frac{\partial y_2}{\partial y_1}\right)\left(\frac{\partial y_1}{\partial W_1}\right)$$

Application of the Chain Rule

Extension to Computational Graphs

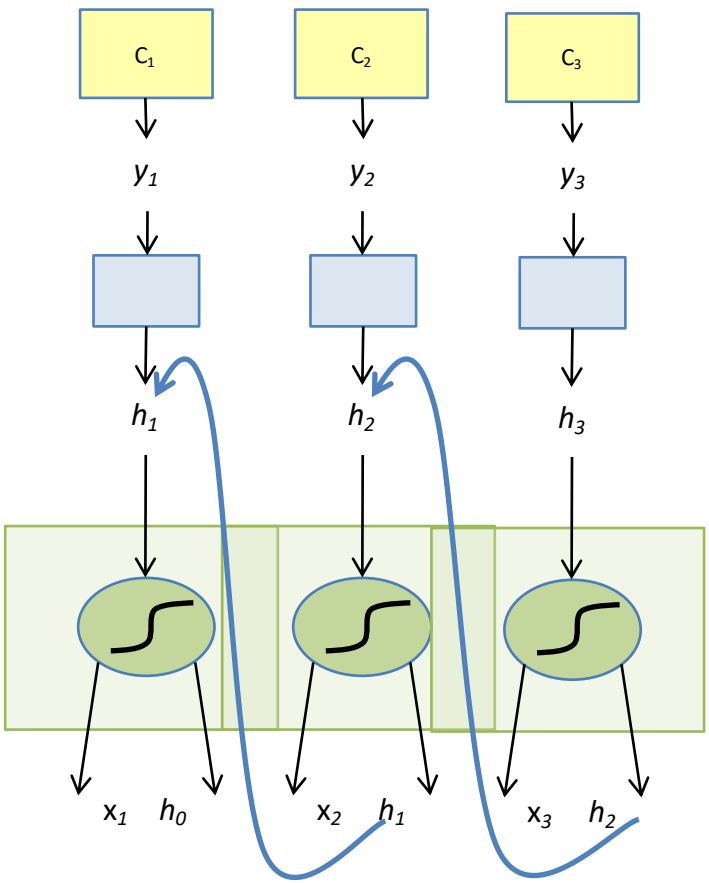


Learning with RNN : Forward pass



- Treat the unfolded network as **one big feed-forward network**!
- RNN takes in **entire sequence as an input**
- Compute gradients using the usual backpropagation : **Back-Propagation Through Time (BPTT)**
- Update **shared weights**

Learning with RNN : Back-Propagation Through Time (BPTT)



$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$$

$$= \tanh\left(\begin{pmatrix} W_{hh} & W_{hx} \end{pmatrix} \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

$$= \tanh\left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

$$\frac{\partial C_t}{\partial h_1} = \left(\frac{\partial C_t}{\partial y_t} \right) \left(\frac{\partial y_t}{\partial h_1} \right)$$

$$= \left(\frac{\partial C_t}{\partial y_t} \right) \left(\frac{\partial y_t}{\partial h_t} \right) \left(\frac{\partial h_t}{\partial h_{t-1}} \right) \cdots \left(\frac{\partial h_2}{\partial h_1} \right)$$

Applications of RNN model

- **Text processing** (language modeling, translation, summarization)
- **Speech** analysis and music
  - Speech recognition
  - music generation
- **Time-series** forecasting
- Sentiment analysis
- Imaging and video analysis:
  - Image captioning
  - Video analysis,
- Bioinformatics sequences (DNA, protein modeling)
- Many more……

Sequential Processing of Non-Sequence Data

- Classify images by taking a series of "glimpses"

Ba, Mnih, and Kavukcuoglu, "Multiple Object Recognition with Visual Attention", ICLR 2015. Gregor et al, "DRAW: A Recurrent Neural Network For Image Generation", ICML 2015 Figure copyright Karol Gregor, Ivo Danihelka, Alex Graves, Danilo Jimenez Rezende, and Daan Wierstra, 2015. Reproduced with permission.

Sequences are everywhere…

FOREIGN MINISTER.

THE SOUND OF

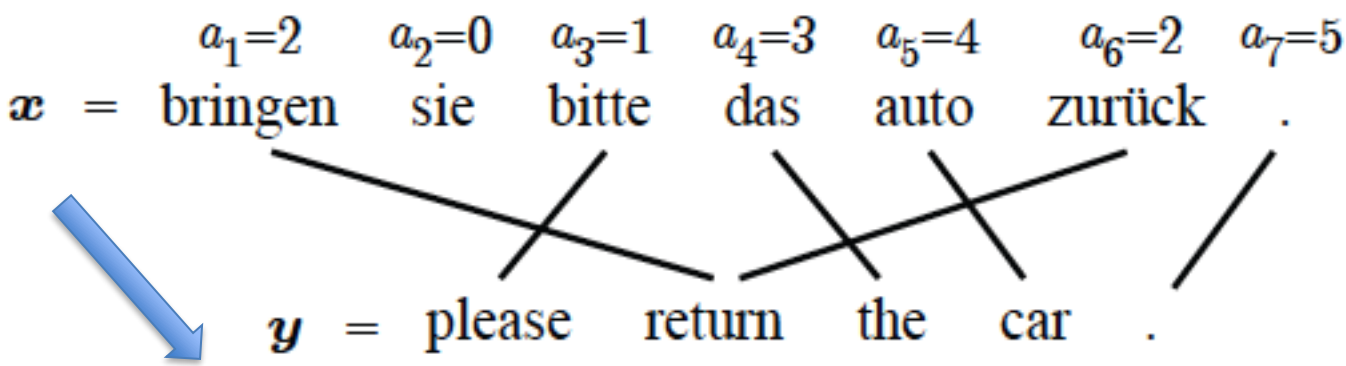
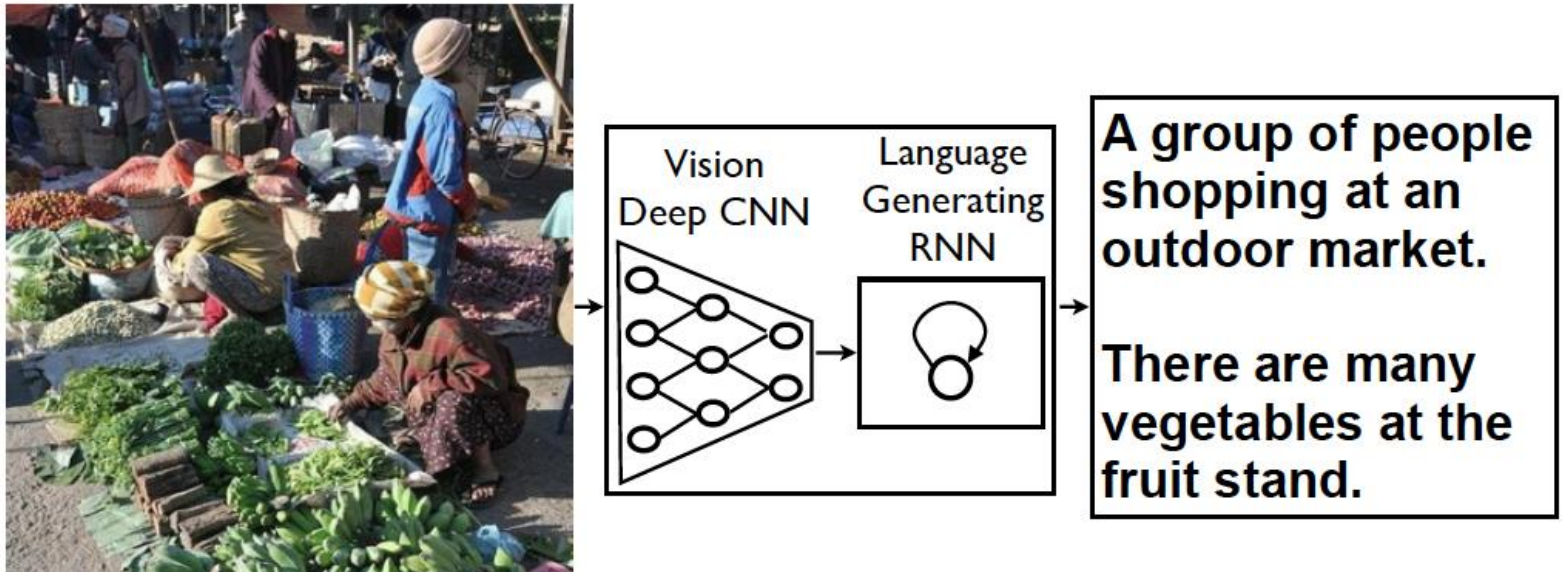
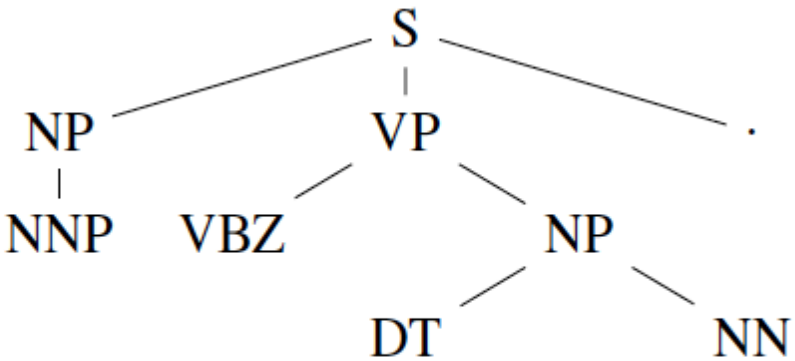


Image Credit: Alex Graves and Kevin Gimpel

Even where you might not expect a sequence…



John has a dog . →



John has a dog . → $(\mathrm{S}\ (\mathrm{NP}\ \mathrm{NNP}\ )_{\mathrm{NP}}\ (\mathrm{VP}\ \mathrm{VBZ}\ (\mathrm{NP}\ \mathrm{DT}\ \mathrm{NN}\ )_{\mathrm{NP}}\ )_{\mathrm{VP}}\ .\ )_{\mathrm{S}}$

Image Credit: Vinyals et al.

Even where you might not expect a sequence…

Classify images by taking a series of “glimpses”

Ba, Mnih, and Kavukcuoglu, “Multiple Object Recognition with Visual Attention”, ICLR 2015.
Gregor et al, “DRAW: A Recurrent Neural Network For Image Generation”, ICML 2015
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RNN Outputs: Language Modeling

VIOLA:

Why, Salisbury must find his flesh and thought

That which I am not aps, not a man and in fire,

To show the reining of the raven and the wars

To grace my hand reproach within, and not a fair are hand,

That Caesar and my goodly father's world;

When I was heaven of presence and our fleets,

We spare with hours, but cut thy council I am great,

Murdered and by thy master's ready there

My power to give thee but so much as hell:

Some service in the noble bondman here,

Would show him to her wine.

KING LEAR:

O, if you were a feeble sight, the courtesy of your law,

Your sight and several breath, will wear the gods

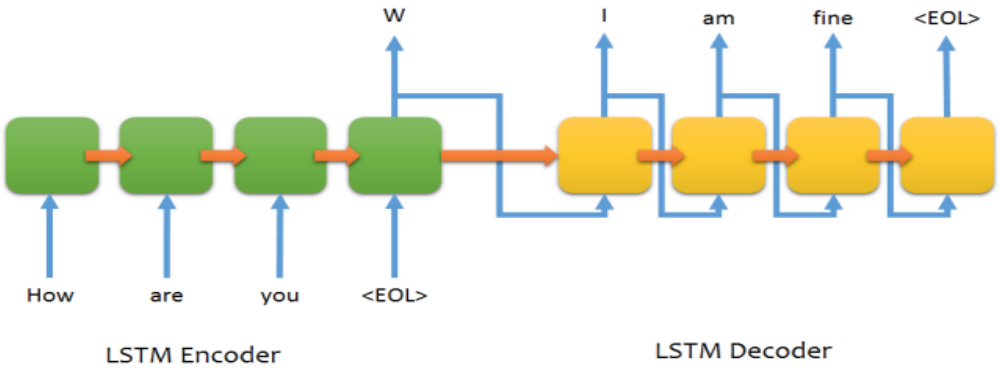
With his heads, and my hands are wonder'd at the deeds,

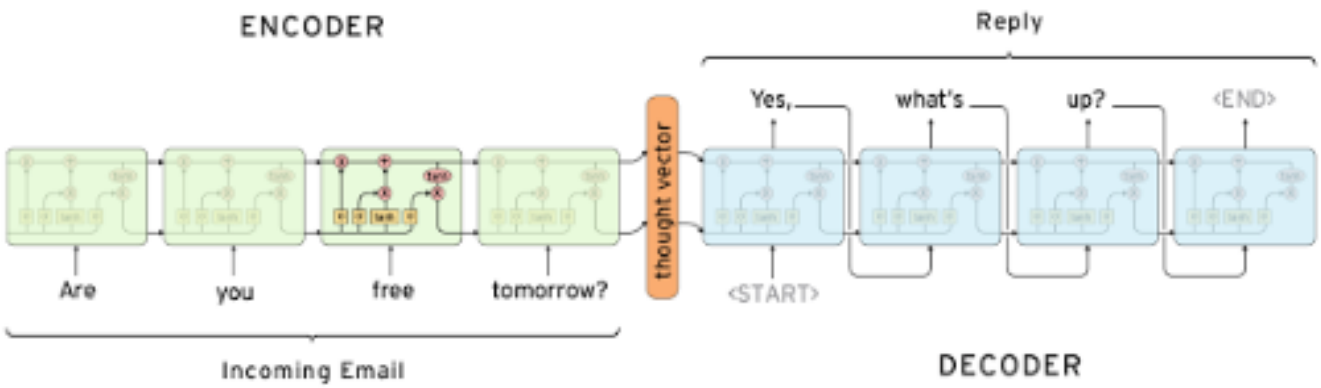
So drop upon your lordship's head, and your opinion

Shall be against your honour.

http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Chatbots



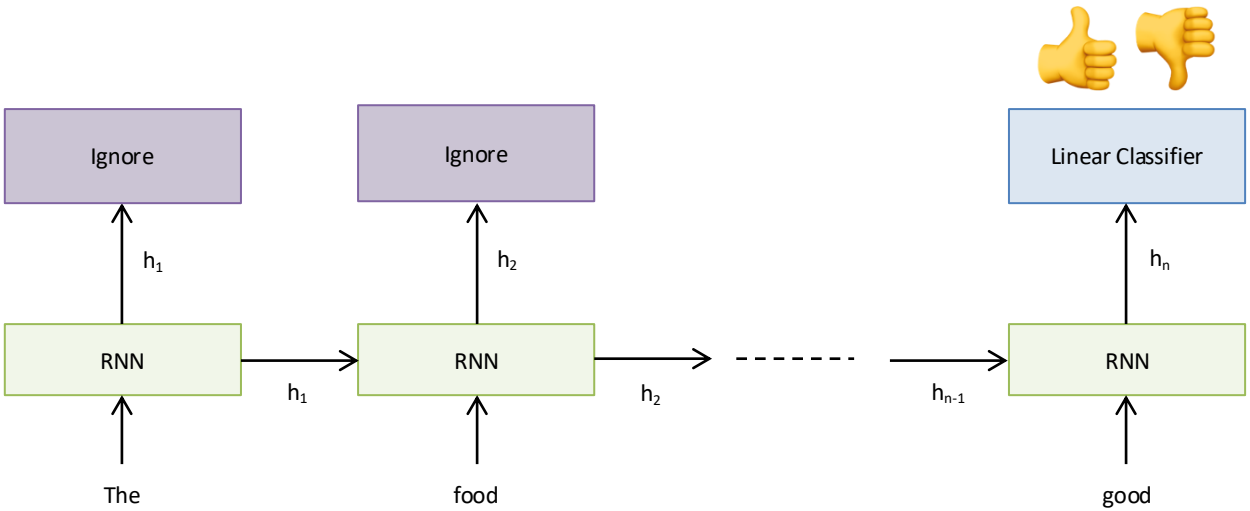


http://suriyadeepan.github.io/2016-06-28-easy-seq2seq/

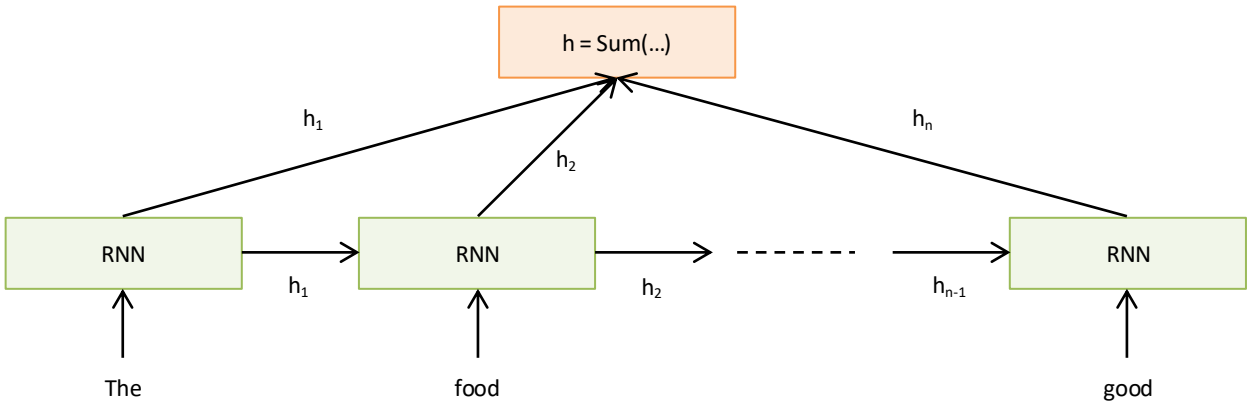
Sentiment analysis

Discovering people opinions, emotions and feelings about a product or service

Sentiment Classification

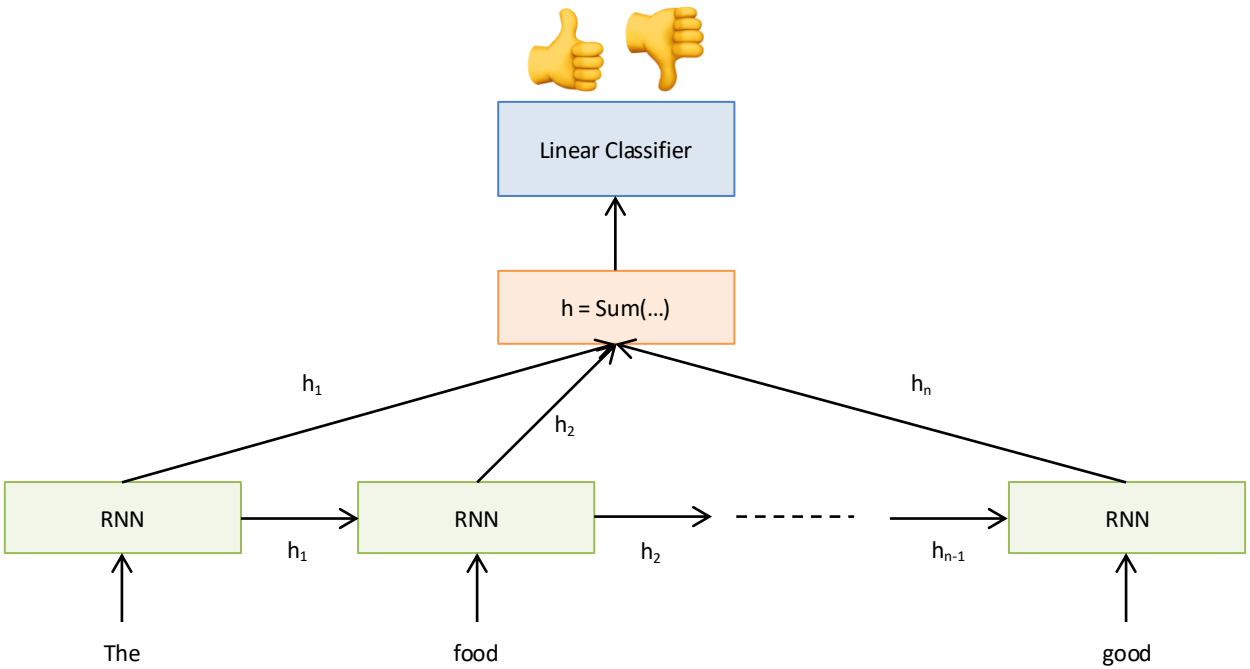


Sentiment Classification



http://deeplearning.net/tutorial/lstm.html

Sentiment Classification



http://deeplearning.net/tutorial/lstm.html

Image Captioning

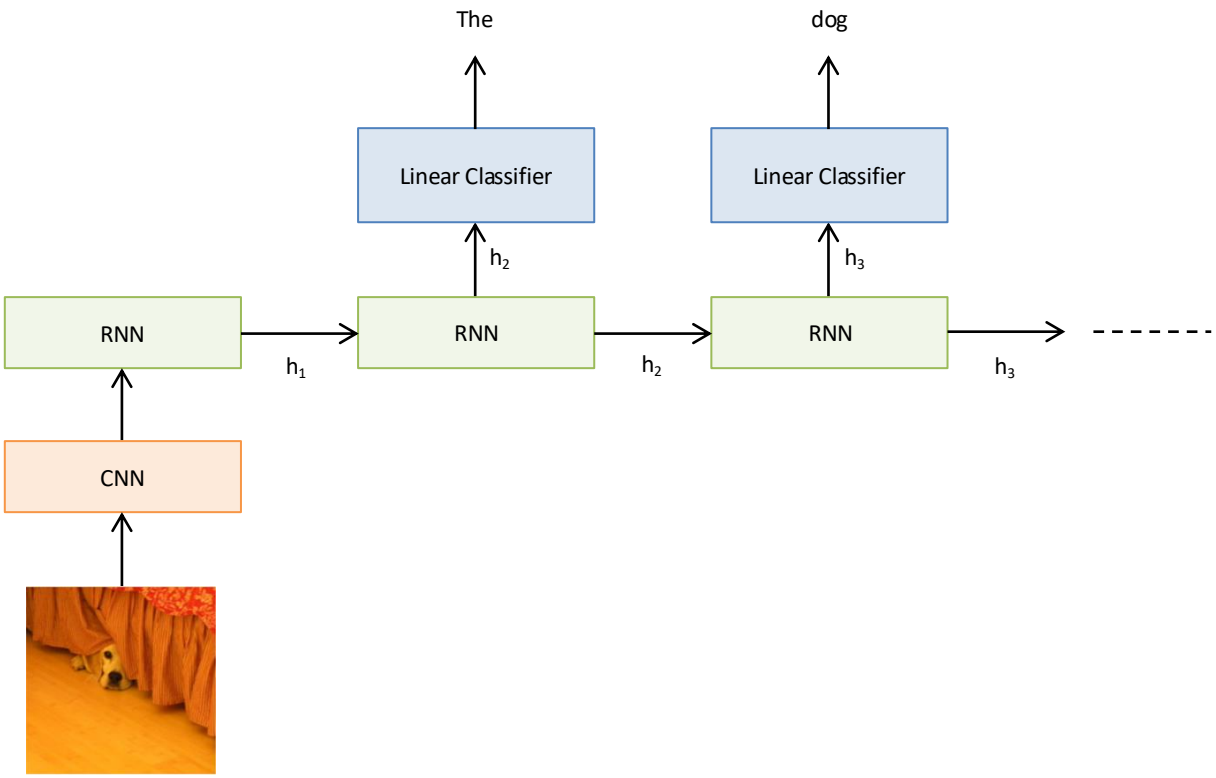
- Given an image, produce a sentence describing its contents

  - Inputs: Image feature (from a CNN)

  - Outputs: Multiple words (let's consider one sentence)

: The dog is hiding

Image Captioning



RNN Outputs: Image Captions

A person riding a motorcycle on a dirt road.

Two dogs play in the grass.

A herd of elephants walking across a dry grass field.

A group of young people playing a game of frisbee.

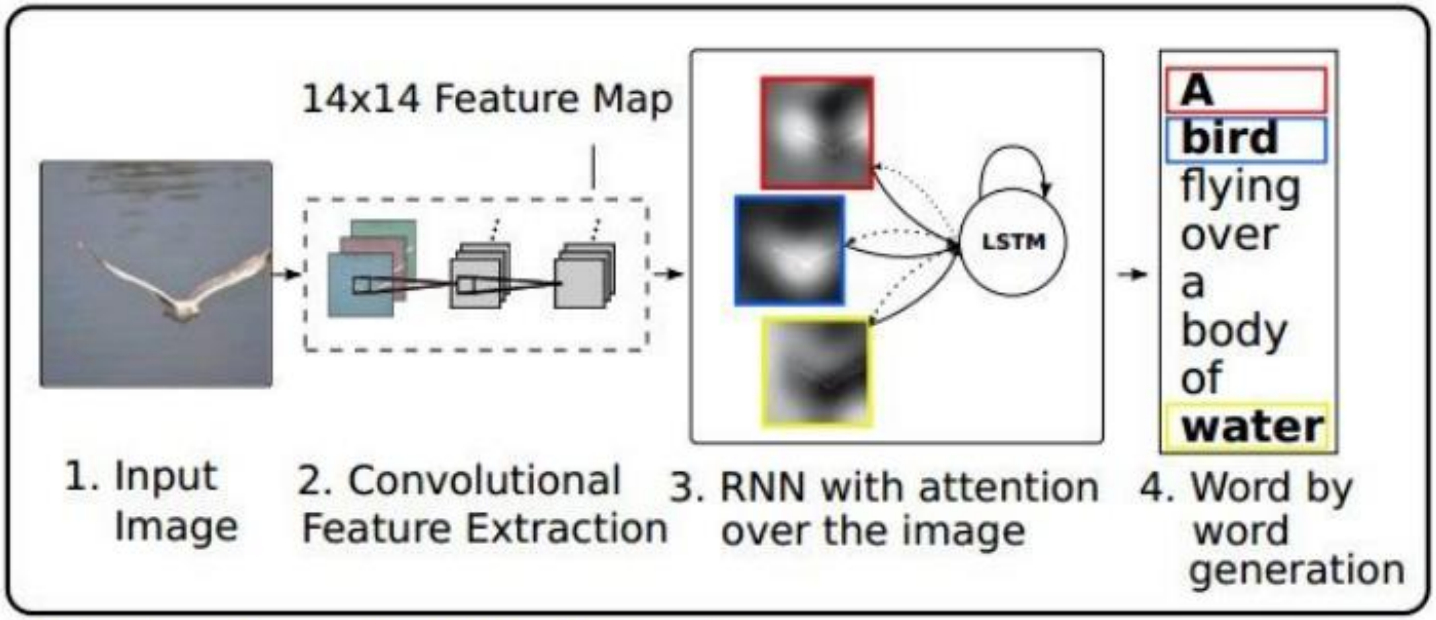
Two hockey players are fighting over the puck.

A close up of a cat laying on a couch.

Show and Tell: A Neural Image Caption Generator, CVPR 15

Image Captioning with Attention

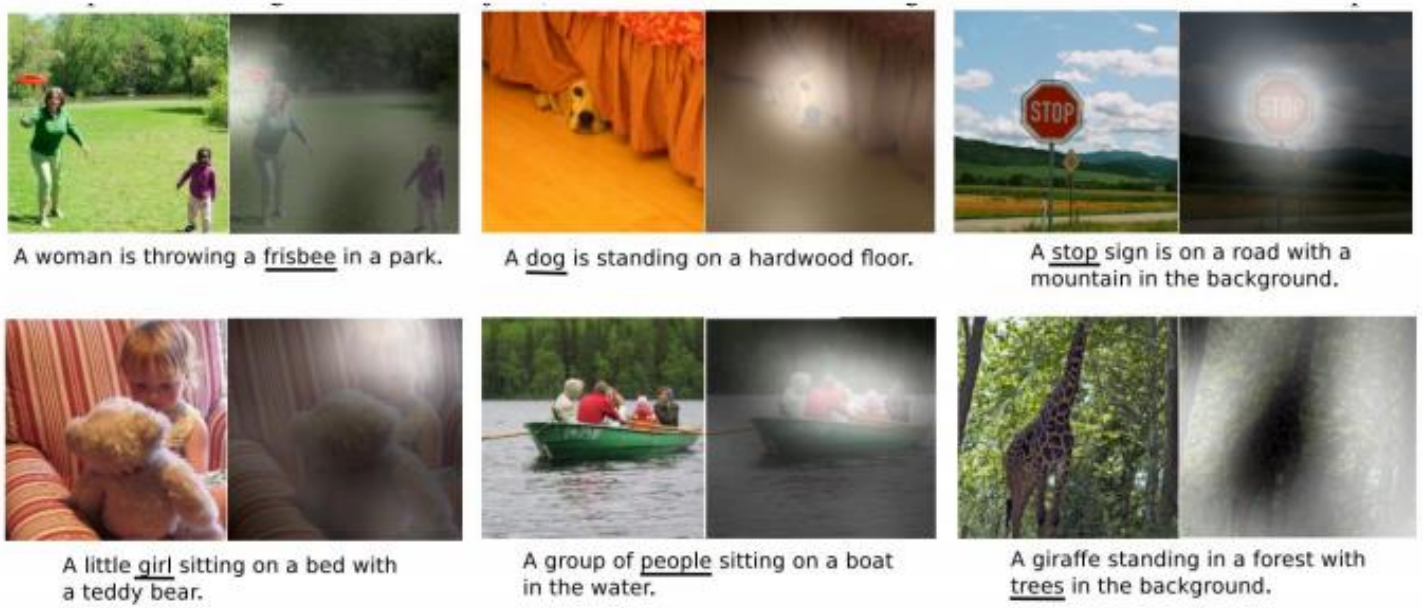
RNN focuses its attention at a different spatial location when generating each word



Xu et al, "Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Figure copyright Kelvin Xu, Jimmy Lei Ba, Jamie Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard S. Zemel, and Yoshua Benchio, 2015. Reproduced with permission.

Image Captioning with Attention



Xu et al, "Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

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Visual Question Answering

Q: What endangered animal is featured on the truck?

A: **A bald eagle.**
A: A sparrow.
A: A humming bird.
A: A raven.

Q: Where will the driver go if turning right?

A: **Onto 24 ¾ Rd.**
A: Onto 25 ¾ Rd.
A: Onto 23 ¾ Rd.
A: Onto Main Street.

Q: When was the picture taken?

A: **During a wedding.**
A: During a bar mitzvah.
A: During a funeral.
A: During a Sunday church service

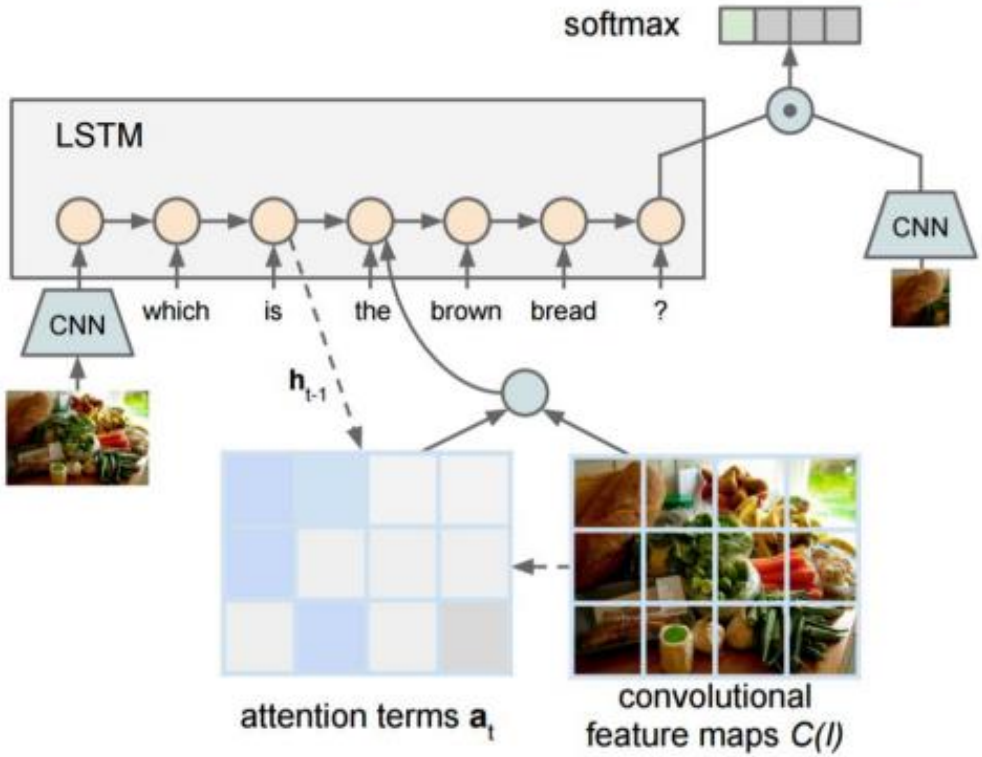
Q: Who is under the umbrella?

A: **Two women.**
A: A child.
A: An old man.
A: A husband and a wife.

Agrawal et al, "VQA: Visual Question Answering", ICCV 2015
Zhu et al, "Visual 7W: Grounded Question Answering in Images", CVPR 2016
Figure from Zhu et al, copyright IEEE 2016. Reproduced for educational purposes.

From: Fei-Fei Li & Justin Johnson & Serena Yeung lecture

Visual Question Answering: RNNs with Attention



What kind of animal is in the photo?
A **cat**.

Why is the person holding a knife?
To cut the **cake** with.

Zhu et al, "Visual 7W: Grounded Question Answering in Images", CVPR 2016
Figures from Zhu et al, copyright IEEE 2016. Reproduced for educational purposes.

From: Fei-Fei Li & Justin Johnson & Serena Yeung lecture

Summary

- Recurrent Neural Networks (RNN)
  - RNNs allow for processing of variable length inputs and outputs by **maintaining state information across time steps**
  - **Various Input-Output scenarios** are possible (Single/Multiple)
- Types of RNNs
- Computational graph for RNN models
- Applications of RNN

- What's next?
  - Long Short-Term Memory (LSTMs)
  - Gated Recurrent Units (GRUs)